

DT1D-Adapter: Lightweight Axial Filtering for Parameter-Efficient Fine-Tuning of Visual Backbones

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Abstract

Pretrained vision backbones are becoming increasingly large, making full fine-tuning expensive in training time, memory usage, and per-task storage, especially when a single backbone must be adapted to multiple tasks or data domains. In the setting of parameter-efficient fine-tuning (PEFT), we propose **DT1D-Adapter**, a lightweight spatial adapter for both convolutional networks (ConvNets) and Transformer-based vision models. The proposed module parameterizes each group of axial depthwise kernels using a group-shared symmetric base kernel on radii $\{1, 2, 4\}$ together with a normalized radius-4 channel-dependent correction. A learned bounded symmetric coefficient shift with displacement $p = 2$ is fused into each axial kernel, producing an effective 13-tap filter; the shifted height- and width-axis kernels are then constrained jointly in ℓ_1 and applied by one depthwise convolution per axis before learned residual injection, providing an explicit spatial inductive bias with a small task-specific parameter budget. The currently completed image-classification experiments show that **DT1D-Adapter** provides a useful accuracy-parameter trade-off in selected transfer settings, although it is not uniformly superior in raw accuracy across all datasets and backbones. In the completed PennFudan binary-segmentation rerun with DeepLabV3-MobileNetV3-Large, DT1D-Adapter achieves $46.345 \pm 1.288\%$ foreground IoU and $63.330 \pm 1.197\%$ Dice over three seeds while adding only 792 adapter-specific parameters; LoRA-Conv obtains the strongest supplied PEFT segmentation result at $49.860 \pm 1.334\%$ IoU, whereas full fine-tuning reaches $67.853 \pm 1.604\%$. In the completed PennFudan ViT-B/16 dense-decoder rerun, DT1D-Adapter achieves $64.279 \pm 0.694\%$ foreground IoU and $78.255 \pm 0.513\%$

Dice with 1,252 trainable parameters, while BitFit reaches the strongest supplied mean at $66.874 \pm 1.558\%$ IoU; the supplied Full fine-tuning run reaches $39.001 \pm 0.991\%$ IoU. In the completed Oxford-IIIT Pet ViT-B/16 semantic-segmentation rerun, DT1D-Adapter achieves $73.634 \pm 0.342\%$ mIoU and $83.058 \pm 0.352\%$ Dice with 2,790 trainable parameters, of which 483 are adapter-specific; BitFit obtains the strongest supplied mean mIoU at $76.671 \pm 0.246\%$. In the updated 10-epoch Oxford-IIIT Pet object-detection rerun with Faster R-CNN/MobileNetV3-FPN, DT1D-Adapter achieves $96.940 \pm 0.359\%$ AP_{50} over three seeds while adding only 792 adapter-specific parameters beyond the trainable detector components. Other figure reruns remain pending in this working manuscript version. Code, configuration files, and reproducibility instructions are publicly available at <https://github.com/tydeptrai21042004/1D-DT-ADAPTER>.

Keywords: parameter-efficient fine-tuning; visual adapter; axial filtering; vision transformer; lightweight visual computing; ConvNet.

1 Introduction

Pretrained vision models are increasingly used as backbones for downstream tasks such as image classification, recognition, detection, segmentation, medical-image analysis, and dense prediction. However, full fine-tuning updates all backbone parameters and can be costly in training resources, memory consumption, and per-task storage, especially when one pretrained model must be adapted to many datasets or domains. This motivates parameter-efficient fine-tuning (PEFT), where the backbone is kept mostly or entirely frozen and only a small set of task-specific parameters is learned [12, 13, 15, 16]. Efficiency can also be pursued from a complementary data-centric perspective. Recent dataset-distillation research uses a noise-unconstrained generative model to construct compact synthetic training sets [1]. Whereas dataset distillation aims to reduce the amount of data required for training, PEFT reduces the number of task-specific parameters that must be optimized and stored. These directions therefore address different, but complementary, components of efficient visual learning.

This adaptation problem is not confined to classification benchmarks. In practical visual-computing systems, a common pretrained backbone may be reused across multiple domains and tasks, including image enhancement and content processing, industrial inspection, segmentation and detection, medical-image analysis, and edge deployment. As the number of task-specific deployments grows, the cost of storing and maintaining a separate fully tuned backbone also grows. These workloads additionally depend on local and directional visual structure, which motivates lightweight adaptation mechanisms that operate directly on spatial feature representations rather than only on global feature statistics. The same requirement arises for Transformer backbones once patch tokens are arranged on their underlying two-dimensional grid. Medical foundation-model applications provide one concrete instance of this broader setting [5, 6].

Against this background, we consider a transfer setting in which a frozen pretrained backbone f_θ is adapted by optimizing a compact set of trainable parameters ϕ . Existing

PEFT methods include prompt tuning, adapter-based tuning, low-rank updates, bias tuning, feature modulation, and convolutional or residual adaptation branches. Many of these methods, however, are tied to particular backbone structures or do not explicitly model spatial filtering.

We therefore propose **DT1D-Adapter**, a structured axial-filtering module for visual PEFT. For each spatial axis and channel group, the adapter represents the direct axial kernel by a group-shared symmetric base component on radii $\{1, 2, 4\}$ together with a low-dimensional channel-dependent correction. A learned bounded symmetric coefficient shift with displacement $p = 2$ is fused into the direct kernel, yielding an effective 13-tap kernel for each enabled axis. The shifted height- and width-axis kernels are jointly constrained in ℓ_1 before one depthwise convolution per axis, after which a learned residual gate injects the task-specific correction into the frozen representation. Because the operator is defined on spatial feature tensors, the same formulation can be applied directly to ConvNet feature maps and to Vision Transformer patch tokens after reshaping them to a two-dimensional grid.

The contributions of this work are threefold.

1. We introduce **DT1D-Adapter**, a lightweight spatial PEFT module that provides a common adaptation interface for ConvNet feature maps and spatially reshaped Vision Transformer patch tokens.
2. We formulate a structured axial kernel in which a group-shared symmetric base response is augmented by a low-dimensional channel-dependent correction and a learned bounded $p = 2$ symmetric coefficient shift. The shift is fused into an effective 13-tap axial kernel, followed by a post-shift joint height-width ℓ_1 constraint and gated residual injection. We characterize its parameter efficiency, the shared/detail separation, the spectral effect of the weighted shift, and its bounded perturbation behavior.
3. We evaluate DT1D-Adapter against representative PEFT baselines and report predictive performance together with trainable-parameter count, storage cost, computational cost, latency, training time, and memory measurements where available.

Our goal is to demonstrate a practical accuracy-parameter trade-off rather than universal superiority over all PEFT methods.

The remainder of this paper is organized as follows. Section 2 reviews related work on PEFT and efficient visual adaptation. Section 3 presents the structural and mathematical characterization of the proposed structured axial operator. Section 4 describes the DT1D-Adapter architecture, backbone insertion strategy, and computational cost. Section 5 reports the experimental results and efficiency-related comparisons. The concluding section summarizes the findings, limitations, and future directions.

Figure 1 shows the current F01 convergence trace for the selected validation-only configuration.

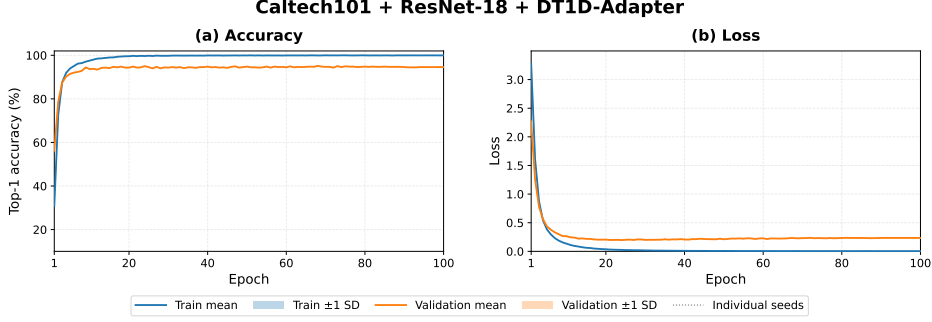


Fig. 1: Convergence behavior of DT1D-Adapter on Caltech101 [64] with an ImageNet-pretrained ResNet-18 backbone for the current F01 seed-0 rerun. The learning rate is selected by validation-only search over $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 5 \times 10^{-3}\}$; the selected learning rate is 3×10^{-4} . The best validation Acc@1 is 95.161% at recorded epoch 67, and the corresponding one-time test evaluation gives 93.541% Acc@1.

2 Related Work

Parameter-efficient fine-tuning (PEFT) in vision can be broadly grouped into prompt tuning, adapter-based tuning, low-rank or partial-parameter tuning, feature modulation, and convolutional or residual adaptation branches. This section reviews these directions with emphasis on their mechanisms, strengths, and limitations, and positions **DT1D-Adapter** relative to convolutional, residual, and axis-aware lightweight adaptation methods.

Prompt tuning learns task-specific tokens while keeping the pretrained backbone frozen. Visual Prompt Tuning (VPT) [15] is a representative approach for Vision Transformers and is attractive because it introduces only a small number of trainable parameters. Neural prompt search methods such as NOAH [55] further study automated prompt and adaptation design for Transformer-based vision models. However, prompt tuning is naturally tied to token-based representations and is less direct for convolutional feature maps, where spatial information is represented as dense $H \times W$ tensors. This motivates adapter designs that can operate directly on spatial feature maps and are not restricted to Transformer token sequences.

Adapter-based tuning inserts small trainable modules into a frozen backbone. Classical adapters were introduced as compact bottleneck modules [12], and vision-oriented variants such as AdaptFormer [17], FacT [56], Lv-Adapter [46], CLIP-Adapter [47], and meta-regularized PEFT [48] study different lightweight adaptation mechanisms for Transformer or vision-language backbones. These methods demonstrate the effectiveness of adding compact trainable components to pretrained models, but many of them are primarily designed around Transformer block structures. By contrast, DT1D-Adapter is formulated on spatial feature tensors and can therefore be inserted into both ConvNet and ViT backbones after reshaping token features into spatial maps.

Vision-specific adapters introduce stronger spatial inductive bias. ConvPass [18] adds convolutional components to ViT adaptation, while RepAdapter [19] uses structural re-parameterization to reduce inference-time overhead. ViT-Adapter [57] further

shows that adapter-style spatial priors can help Transformer backbones serve dense vision tasks, and structured pruning adapters [45] provide another route for reducing adaptation cost. These works support the importance of spatial structure in visual adaptation. However, convolution-based branches may still introduce non-negligible parameter or computational cost depending on the kernel size, channel mixing, and insertion strategy. DT1D-Adapter follows this spatial-adaptation direction but restricts the learned spatial operator to group-shared symmetric axial base kernels with a low-dimensional channel-dependent correction, thereby avoiding a fully parameterized two-dimensional adaptation branch.

Low-rank and partial-parameter methods provide another lightweight PEFT route. LoRA [13] learns low-rank weight updates, and later variants such as LoRand [58] further explore low-rank randomized adaptation. Recent visual low-rank adaptation studies, such as HLoRA-MER [70], further show that high-level LoRA-style adaptation can be effective for visual recognition while maintaining a small trainable-parameter budget. BitFit [14] updates only bias parameters, SSF [16] learns feature-wise scale and shift parameters, and MoNA [59] represents a recent vision-oriented PEFT direction. These methods are simple and storage-efficient, but they mainly modify parameter subspaces or feature statistics rather than explicitly modeling local or directional spatial structure. Therefore, they are complementary to DT1D-Adapter, which introduces a restricted structured axial filtering branch while keeping the backbone frozen.

Conv-Adapter [36] and Residual Adapters [38] are among the closest baselines to our work. Conv-Adapter introduces convolutional adaptation modules for ConvNet backbones, while Residual Adapters use residual branches to adapt a shared backbone across domains. Both lines demonstrate the usefulness of residual adaptation with visual inductive bias. However, their parameter cost can grow with the branch design, the number of inserted layers, and the amount of channel mixing. DT1D-Adapter follows the residual-adaptation principle but constrains the spatial branch through a group-shared symmetric base response, a low-dimensional channel-dependent correction, a learned bounded symmetric coefficient shift fused into each axial kernel, post-shift joint height-width normalization, and a scalar residual gate.

Attention-style and axis-aware lightweight modules are also relevant. Squeeze-and-Excitation Networks [44] and BAM [42] show that lightweight channel or channel-spatial recalibration can improve convolutional representations. Axial Attention [49] and Axial-DeepLab [50] show that processing visual representations along separate spatial axes can reduce the cost of multidimensional modeling, while Coordinate Attention [51] encodes direction-aware spatial information through separated height and width contexts. Large-kernel and depthwise-convolutional designs such as ConvNeXt [53] and RepLKNet [54] also support the usefulness of efficient spatial filtering in modern vision backbones. Efficient multi-scale processing is further explored by EAPT, which introduces an attention-pyramid Transformer for image-processing tasks [2]. Although EAPT is a complete visual architecture rather than a PEFT module, its design provides relevant context for efficient Transformer-based spatial representation learning. Taken together, these studies motivate axis-aware and depthwise spatial operations, while DT1D-Adapter specializes these principles to parameter-efficient fine-tuning through a compact structured axial-filtering branch.

Recent studies on efficient visual adaptation and lightweight model design further indicate the importance of reducing adaptation cost while preserving transfer performance. Examples include parameter-diminish fine-tuning for Transformer-based models [39], lightweight CNN-ViT designs [40], and environment-aware domain adaptation [41]. For dense medical-image prediction, Wen et al. combine depthwise separable convolution with frequency-domain enhancement in a lightweight retinal-vessel segmentation network [3]. Their design illustrates how compact spatial operators can support dense prediction while limiting model complexity. DT1D-Adapter addresses a complementary setting, namely the parameter-efficient adaptation of pretrained backbones, but shares the broader objective of introducing effective spatial modeling with a controlled task-specific parameter budget. A frozen-backbone RGB-IR detection framework with multi-receptive-field attention [71] provides another related example by combining fixed pretrained representations with lightweight task-specific adaptation and multi-scale receptive-field modeling. Collectively, these studies reinforce the broader trend toward efficient adaptation, compact spatial modeling, and reduced task-specific deployment cost.

Within this landscape, the individual operations used by DT1D-Adapter—axial filtering, depthwise convolution, symmetric kernels, shift operators, residual adaptation, and coefficient normalization—are established techniques. The distinction lies in how they are composed into a restricted task-specific spatial kernel family. Rather than learning an independent axial kernel for every channel, DT1D-Adapter shares a symmetric base kernel within each channel group, introduces only a low-dimensional channel-dependent correction, and learns a bounded axis-specific symmetric coefficient shift that is fused into the final axial kernel. This differs from axial attention and Coordinate Attention [49, 51], which construct content-dependent attention or directional context, whereas DT1D-Adapter remains a convolutional operator after training. The final formulation uses one finite-support kernel per enabled axis rather than evaluating several separate dilated-convolution branches [29], and no input-dependent branch selection is used as in selective-kernel mechanisms [61]. Compared with more general convolutional adapters, the proposed branch therefore restricts the learned spatial kernel family instead of introducing a freely parameterized two-dimensional spatial or channel-mixing module. Post-shift joint control of the height- and width-axis kernel magnitudes and gated residual injection complete the same compact spatial adaptation interface for ConvNet feature maps and spatially arranged ViT patch representations.

Table 1 summarizes the position of DT1D-Adapter relative to representative PEFT and lightweight visual adaptation methods.

The effectiveness of PEFT depends not only on the adaptation module, but also on its insertion location and parameter budget. This is important in vision backbones because early, middle, and deep layers represent different levels of texture, parts, and semantics. Therefore, PEFT methods should be evaluated with respect to accuracy, placement sensitivity, parameter budget, and backbone type.

SPT [22] studies sensitivity-based criteria for selecting task-relevant tuning locations and allocating parameters under a fixed budget. V-PETL [23] further shows that placement and design choices can strongly affect transfer performance. Large-scale empirical studies also suggest that hyperparameter settings, training protocol, and

Table 1: Positioning of DT1D-Adapter relative to representative PEFT and lightweight visual adaptation methods.

Method family	Main mechanism	Main strength	Relation to DT1D-Adapter
VPT [15]	Learnable prompt tokens for ViTs	Very small trainable-parameter budget	Token-based design; less direct for ConvNet feature maps.
AdaptFormer [17]	Bottleneck adapter inside Transformer blocks	Effective ViT adaptation	Mainly designed for Transformer block structures.
LoRA [13]	Low-rank weight updates	General and parameter-efficient	Adapts weights but does not explicitly model spatial filtering.
BitFit / SSF [14, 16]	Bias tuning or feature-wise scale-shift tuning	Extremely lightweight	Modulates parameters or features without an explicit axial spatial prior.
ConvPass / Conv-Adapter [18, 36]	Convolutional adaptation branch	Strong visual inductive bias	Spatially expressive, but parameter cost can increase with branch design.
Residual Adapters [38]	Residual task- or domain-specific branches	Effective multi-domain adaptation	Related residual principle, but branch complexity may increase storage cost.
BAM-Tuning [42]	Channel-spatial attention branch	Lightweight feature recalibration	Attention-based rather than structured axial-filtering based.
DT1D-Adapter	Group-shared symmetric axial base and channel correction with fused bounded weighted shift	Restricted spatial kernel family with controlled task-specific parameter cost	One finite-support axial kernel per enabled axis; spatial PEFT interface for ConvNet maps and reshaped ViT patch grids.

parameter-budget alignment can change the ranking of PEFT methods [24]. These findings motivate our later discussion of insertion strategy and hyperparameter selection for DT1D-Adapter.

Fair evaluation is also important because PEFT methods differ in trainable parameters, storage cost, training memory, and inference overhead. Thus, accuracy alone can be misleading. V-PETL Bench [25] provides a unified benchmark for vision PEFT, while

recent surveys summarize evaluation settings and open challenges [26, 27]. These studies motivate our experimental protocol, which reports accuracy, trainable parameters, storage cost, and computational or memory-related measurements where available.

3 Mathematical Characterization and Design Rationale

This section characterizes the DT1D-Adapter parameterization used by the final implementation and relates its mathematical structure to the design choices used in the code. Standard finite-support convolution and Fourier analysis [62, 63] are used only as analytical tools to describe the restricted kernel family, the spectral effect of the learned weighted shift, and the post-shift joint kernel constraint.

3.1 Structured shared and channel-dependent axial kernel

Let

$$x \in \mathbb{R}^{B \times C \times H \times W} \quad (1)$$

be an intermediate feature tensor, and let

$$\mathcal{A} \subseteq \{h, w\} \quad (2)$$

denote the enabled spatial axes. Channels are divided into groups of size g_{ch} , with

$$G = \left\lceil \frac{C}{g_{\text{ch}}} \right\rceil. \quad (3)$$

For each axis $u \in \mathcal{A}$ and channel group q , DT1D-Adapter learns a symmetric group-shared base kernel over the fixed radii

$$\mathcal{R} = \{1, 2, 4\}. \quad (4)$$

Writing δ_r for the unit impulse at offset r , the base component is

$$b_{u,q} = \beta_{u,q,0}\delta_0 + \sum_{r \in \mathcal{R}} \beta_{u,q,r} (\delta_{-r} + \delta_r). \quad (5)$$

Thus, the shared base response uses four learned coefficients per axis and group and occupies a direct 9-tap support from -4 to $+4$.

To allow limited channel-specific variation without learning an independent spatial kernel for every channel, DT1D-Adapter adds one channel-dependent correction component. Define the normalized symmetric second-difference basis function at radius 4

$$\psi_4 = \frac{\delta_{-4} - 2\delta_0 + \delta_4}{\sqrt{6}}, \quad \sum_r \psi_4[r] = 0, \quad \|\psi_4\|_2 = 1. \quad (6)$$

Within every non-degenerate group, a fixed zero-mean within-group modulation vector a_q is constructed so that

$$\mathbf{1}^\top a_q = 0, \quad \|a_q\|_2 = 1. \quad (7)$$

If channel c belongs to group $g(c)$, its direct pre-shift axial kernel is

$$k_{u,c} = b_{u,g(c)} + \eta_{u,g(c)} a_c \psi_4, \quad (8)$$

where $\eta_{u,g(c)}$ is a learned group-level correction amplitude. For a singleton remainder group, the zero-mean modulation is inactive, so the kernel reduces to the shared base component.

3.2 Parameter efficiency and separation of shared and channel-dependent components

The structured parameterization reduces the learned spatial degrees of freedom at the group level.

Proposition 1 (Spatial degrees of freedom) *Consider a group containing $n \geq 2$ channels. An unrestricted 13-tap depthwise axial filter has $13n$ free spatial coefficients for that group and axis. A per-channel symmetric kernel restricted to the coordinates $\{0, \pm 1, \pm 2, \pm 4\}$ has $4n$ free coefficients. By contrast, the direct DT1D kernel family in (8) uses four group-shared base coefficients and one group-level correction amplitude, hence five learned spatial coefficients per group and axis before the weighted-shift scalar is applied.*

Proof The base kernel in (5) belongs to the four-dimensional span

$$\text{span}\{\delta_0, \delta_{-1} + \delta_1, \delta_{-2} + \delta_2, \delta_{-4} + \delta_4\}.$$

The correction term adds one learned scalar coefficient $\eta_{u,q}$ along the fixed channel-spatial direction $a_q \psi_4^\top$. Since a_q is nonzero and centered, this direction is linearly independent of the group-common base-kernel directions. Therefore the learned direct group-level spatial parameterization has five degrees of freedom. $\square$

With $A = |\mathcal{A}|$ enabled axes, the direct shared/detail coefficients contribute

$$N_{\text{direct}} = 5AG, \quad G = \left\lceil \frac{C}{g_{\text{ch}}} \right\rceil, \quad (9)$$

followed by A learned axis-specific weighted-shift scalars and one learned residual gate in the canonical configuration. The reduction arises from restricting the channel-spatial kernel family rather than from merely replacing one convolution by another.

The same construction separates the common group response from the channel-dependent correction. For one group and axis, write the direct kernel tensor as

$$K_{u,q} = \mathbf{1} b_{u,q}^\top + \eta_{u,q} a_q \psi_4^\top. \quad (10)$$

Proposition 2 (Pre-shift orthogonality of shared and channel-dependent components) *The group-common base component and the channel-dependent correction component are orthogonal in the channel-spatial Frobenius inner product:*

$$\left\langle \mathbf{1} b_{u,q}^\top, \eta_{u,q} a_q \psi_4^\top \right\rangle_F = 0. \quad (11)$$

Proof Using the separable form of the two terms and (7),

$$\left\langle \mathbf{1} b^\top, \eta a \psi_4^\top \right\rangle_F = \eta (\mathbf{1}^\top a) (b^\top \psi_4) = 0.$$

□

This orthogonality is a property of the direct shared/detail decomposition before the learned weighted shift and before the final per-channel projection. Practically, the shared component models the group-common spatial response, while the channel-dependent correction introduces limited channel variation without allocating a free kernel to every channel.

3.3 Spectral structure, weighted shift, joint normalization, and residual control

Because the direct base kernel is symmetric, its one-dimensional Fourier response is real and even:

$$\widehat{b}_{u,q}(\omega) = \beta_{u,q,0} + 2 \sum_{r \in \{1,2,4\}} \beta_{u,q,r} \cos(r\omega). \quad (12)$$

The radius-4 correction basis function has response

$$\widehat{\psi}_4(\omega) = \frac{2 \cos(4\omega) - 2}{\sqrt{6}} = -\frac{4}{\sqrt{6}} \sin^2(2\omega), \quad (13)$$

and therefore $\widehat{\psi}_4(0) = 0$.

The final proposal then applies a learned symmetric coefficient shift to each enabled axis. For displacement $p = 2$, let $S_{\pm 2}$ denote a two-sample shift of a finite kernel on the expanded support and define

$$\bar{k}_{u,c} = \mathcal{W}_{2,\lambda_u} k_{u,c} = (1 - \lambda_u) k_{u,c} + \frac{\lambda_u}{2} (S_{-2} k_{u,c} + S_{+2} k_{u,c}), \quad (14)$$

where

$$\lambda_u = 0.5 \tanh(\theta_u). \quad (15)$$

The height and width axes use independent learned scalars. They are initialized at $\lambda_u = 0$, so training starts from the direct kernel in (8). The direct radius is four and the shift displacement is two, giving an effective radius of six and an effective 13-tap kernel. The shift is fused into the kernel coefficients and therefore does not create an additional convolution branch.

For a finite kernel k with Fourier response $\widehat{k}(\omega)$, the weighted shift has response

$$\widehat{\mathcal{W}_{2,\lambda}k}(\omega) = [1 - \lambda + \lambda \cos(2\omega)] \widehat{k}(\omega). \quad (16)$$

Thus, the learned scalar gives a bounded axis-specific spectral reweighting of the structured kernel rather than a content-dependent routing mechanism. Figure 2 illustrates the compact direct spectral response and the $p = 2$ weighted-shift multiplier. The plotted nonzero λ is illustrative; the learned shift is initialized at zero in the actual model.

After the weighted shift, the implementation jointly projects the height and width kernels for each channel. Define

$$\rho_c = \max\{1, \|\bar{k}_{h,c}\|_1 + \|\bar{k}_{w,c}\|_1\}, \quad (17)$$

with the absent axis omitted in a single-axis configuration, and set

$$\tilde{k}_{u,c} = \frac{\bar{k}_{u,c}}{\rho_c}. \quad (18)$$

Then

$$\|\tilde{k}_{h,c}\|_1 + \|\tilde{k}_{w,c}\|_1 \leq 1. \quad (19)$$

Proposition 3 (Bounded axial response under translation-invariant or zero-extension convolution) *Let*

$$\mathcal{T}(x)_c = \tilde{k}_{h,c} * x_c + \tilde{k}_{w,c} * x_c. \quad (20)$$

Under translation-invariant convolution or zero extension, for every $1 \leq p \leq \infty$,

$$\|\mathcal{T}(x)\|_{\ell_p} \leq \|x\|_{\ell_p}. \quad (21)$$

Proof Young’s convolution inequality gives

$$\|\tilde{k}_{h,c} * x_c\|_{\ell_p} \leq \|\tilde{k}_{h,c}\|_1 \|x_c\|_{\ell_p}$$

and the analogous bound for the width axis. Combining the two estimates with (19) gives the result channel-wise and hence for the complete tensor. $\square$

The implementation uses replicate padding as its finite-grid boundary treatment. We therefore use Proposition 3 to motivate the coefficient-mass projection itself and do not claim a general ℓ_p contraction for replicate padding. The corresponding pointwise/ ℓ_∞ coefficient-mass control remains immediate from (19).

Finally, the adapter output is

$$\text{DT1D}(x) = x + s\gamma \mathcal{T}(x), \quad (22)$$

where $s = 1$ and γ is a learned scalar gate initialized at 0.01. Under the same convolution assumptions,

$$\|\text{DT1D}(x) - x\|_{\ell_p} \leq |s\gamma| \|x\|_{\ell_p}. \quad (23)$$

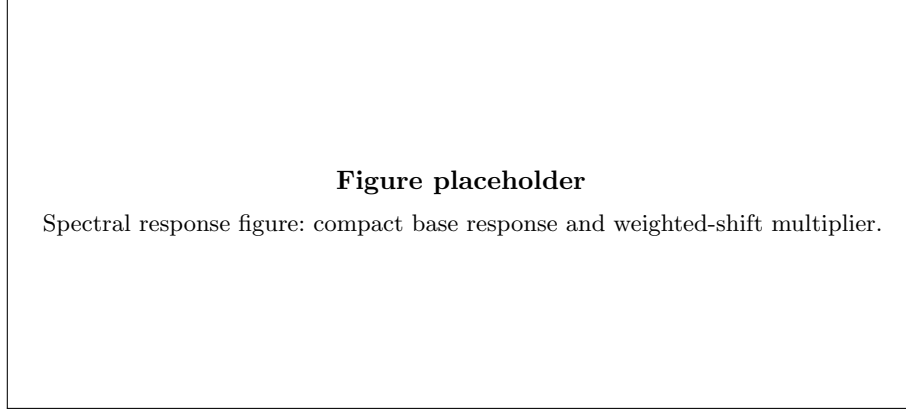


Fig. 2: Spectral interpretation of the DT1D-Adapter parameterization. Placeholder reserved for the compact symmetric base response and the $p = 2$ weighted-shift spectral modulation.

Thus, the branch starts as a small near-identity perturbation, while the learned shift and gate respectively control the shape and magnitude of the structured spatial correction.

4 Proposed Method

This section describes how the structured kernel from Section 3 is used as a trainable residual branch in frozen visual backbones.

4.1 Adapter architecture, axial filtering, and backbone insertion

Given a spatial feature map

$$x \in \mathbb{R}^{B \times C \times H \times W}, \quad (24)$$

DT1D-Adapter partitions the channels into groups of size $g_{\text{ch}} = 16$ and constructs the direct shared/detail kernel in (8) for each enabled spatial axis. It then applies the learned axis-specific weighted shift in (14) and the post-shift joint ℓ_1 projection in (18). For related discrete weighted-convolution constructions, see Definition 1 (Eq. (13)) of Huong and Ty [73], which defines a generalized h -Hartley-cosine convolution with weight $\gamma(y) = \cos(yh)$ on $\mathbb{T}_h$ and is subsequently interpreted as a discrete filtering mechanism. The two-axis response is

$$z = \text{DWConv}_{13 \times 1}(x; \tilde{k}_h) + \text{DWConv}_{1 \times 13}(x; \tilde{k}_w), \quad (25)$$

and the final output is

$$\text{out} = x + s\gamma z, \quad (26)$$

with residual scale $s = 1$ and a learned gate initialized at $\gamma_0 = 0.01$. The branch requires one axial depthwise convolution per enabled axis; the weighted shift is already fused into the corresponding 13-tap kernel. A pointwise channel-mixing block is not used in the canonical proposal.

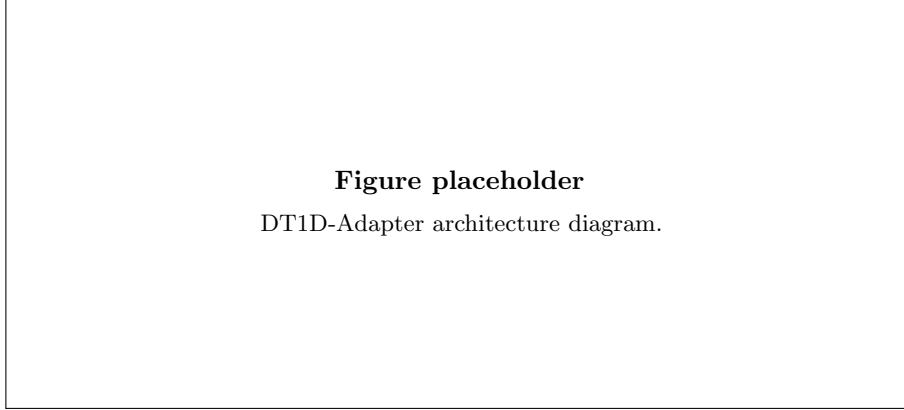


Fig. 3: Architecture of the final DT1D-Adapter data path. Placeholder reserved for compact shared/detail kernel formation, the fused $p = 2$ weighted shift, post-shift joint height–width projection, two axial depthwise convolutions, fusion, and learned residual injection.

Because DT1D-Adapter operates on spatial tensors, it can be inserted into both CNN and Transformer backbones. For CNNs, intermediate activations already have shape $B \times C \times H \times W$, so the adapter can be inserted after selected convolutional blocks or residual stages, in the same general setting as lightweight residual visual adaptation [20, 38]. During fine-tuning, the backbone remains frozen while the DT1D parameters, the two axis-specific shift scalars, the residual gate, and the task-specific head are updated.

For Vision Transformers [11], the patch-token sequence is reshaped to a two-dimensional grid. If $N = H_p W_p$ patch tokens have embedding dimension C , the patch features are represented as

$$x \in \mathbb{R}^{B \times C \times H_p \times W_p}, \quad (27)$$

adapted with the same DT1D operator, and then reshaped back to the token sequence. A class token, when present, is kept separate and concatenated back after the patch tokens are restored. This gives the method a common spatial adaptation interface across ConvNet maps and ViT patch grids.

Figure 3 summarizes the complete data path of the final proposal: compact shared/detail kernel formation, the fused $p = 2$ weighted shift, post-shift joint height–width projection, two axial depthwise convolutions, fusion, and learned residual injection.

4.2 Configuration, computational cost, and algorithm

Unless otherwise stated, DT1D-Adapter uses both spatial axes, channel-group size $g_{\text{ch}} = 16$, direct symmetric radii $\{1, 2, 4\}$, the normalized radius-4 second-difference basis function, learned axis-specific weighted shifts with $p = 2$ and $\lambda_u = 0.5 \tanh \theta_u$ initialized at zero, joint ℓ_1 projection applied after the weighted shift, a learned residual

gate initialized at 0.01, replicate padding, and no pointwise channel-mixing block. The direct kernel length is 9 and the effective shifted kernel length is $K = 13$.

Ignoring the small cost of constructing and fusing the structured kernel coefficients, one depthwise axial convolution costs approximately

$$\mathcal{O}(BCHWK). \quad (28)$$

With $A = |\mathcal{A}|$ enabled axes, the filtering cost is therefore

$$\mathcal{O}(BCHWAK). \quad (29)$$

For the two-axis configuration with $K = 13$, the axial factor is $2K = 26$, compared with $K^2 = 169$ for a full 13×13 depthwise spatial kernel over the same effective support. This comparison describes arithmetic structure only; it does not imply universally lower wall-clock latency, which also depends on tensor shapes, memory traffic, backend implementation, adapter placement, and hardware kernel efficiency.

Algorithm 1 DT1D-Adapter on a feature tensor

Require: Feature tensor x ; group size $g_{\text{ch}} = 16$; shared-base coefficients β ; correction amplitudes η ; shift parameters θ_h, θ_w ; residual gate γ

Ensure: out with the same shape as x

```

1 Partition the  $C$  channels into  $G = \lceil C/g_{\text{ch}} \rceil$  groups
2 for each enabled axis  $u \in \mathcal{A}$  do
3   for each channel group  $q$  do
4     Construct the symmetric base kernel  $b_{u,q}$  on  $\{0, \pm 1, \pm 2, \pm 4\}$ 
5     Form the centered channel-dependent correction  $\eta_{u,q} a_q \psi_4$ 
6     Assign the direct kernel  $k_{u,c} \leftarrow b_{u,q} + \eta_{u,q} a_c \psi_4$  to each channel  $c$  in group  $q$ 
7   end for
8    $\lambda_u \leftarrow 0.5 \tanh(\theta_u)$ 
9   Fuse the  $p = 2$  weighted shift  $\bar{k}_{u,c} \leftarrow (1 - \lambda_u)k_{u,c} + \frac{\lambda_u}{2}(S_{-2}k_{u,c} + S_{+2}k_{u,c})$ 
10 end for
11 Jointly project the shifted height and width kernels using (18)
12  $z_h \leftarrow \text{DWConv}_{13 \times 1}(x; \tilde{k}_h)$ 
13  $z_w \leftarrow \text{DWConv}_{1 \times 13}(x; \tilde{k}_w)$ 
14  $z \leftarrow z_h + z_w$ 
15 out  $\leftarrow x + \gamma z$ 
16 return out
```

5 Experiments

This section reports the results currently available from the supplied rerun archives. To avoid mixing current reruns with legacy values, only cells supported by the present result set are populated. All other numerical cells are intentionally left blank for later

update. Likewise, result figures whose current outputs are not included in the supplied result set are retained as labeled placeholders.

5.1 Experimental protocol

The current result set includes the previously incorporated completed or partially completed reruns together with the newly supplied C10, C11, and C13 CNN result archives and the updated D06/D07 dense-prediction archive. F01 provides a single-seed Caltech101/ResNet-18 100-epoch convergence rerun for DT1D-Adapter; F04 provides a single-seed FGVC-Aircraft/ResNet-18 10-epoch accuracy-parameter trade-off with seven currently completed methods; and C12 provides three-seed Caltech101/ResNet-18 10-epoch comparison results for seven currently completed methods. In F04 and C12, the missing current runs remain unreported rather than being filled from legacy results. First, the core CNN ablation uses DTD/ResNet-18 and Flowers102/ResNet-50 with seeds $\{0, 1, 2\}$. Second, the CNN comparison reruns for DTD/ResNet-50 and Flowers102/ResNet-50 use validation-only learning-rate selection over the common grid $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 5 \times 10^{-3}\}$, followed by test evaluation at the selected best-validation checkpoint. The available CNN comparison archives are partial, so only completed method rows are populated. Third, the supplied ViT-B/16 result sets use a validation-only method-specific learning-rate search on tuning seed 42 followed by final evaluation with seeds $\{0, 1, 2\}$. VTAB-DTD and VTAB-EuroSAT use the VTAB-1k `train800/val200/official-test` protocol, while Flowers102 uses its official train/validation/test split. Fourth, the completed D05 binary-mask segmentation rerun evaluates nine methods on PennFudan using DeepLabV3-MobileNetV3-Large with input size 224×224 , batch size 8, 100 epochs, AdamW, learning rate 10^{-3} , weight decay 10^{-4} , and seeds $\{0, 1, 2\}$. For each segmentation seed, the best checkpoint is selected by validation IoU and the test split is evaluated once. Fifth, the completed D04A/D04B PennFudan ViT-B/16 dense-decoder rerun evaluates DT1D-Adapter, Linear probing, BitFit, and Full fine-tuning with input size 224×224 , batch size 8, 100 epochs, AdamW, learning rate 10^{-3} , weight decay 10^{-4} , cosine learning-rate scheduling, and seeds $\{0, 1, 2\}$. The best checkpoint for each seed is selected by validation IoU, with the test split evaluated once at that checkpoint. Sixth, the updated D06 object-detection rerun evaluates DT1D-Adapter, Full fine-tuning, the task-head-only baseline, Conv-Adapter, and BAM on Oxford-IIIT Pet detection with Faster R-CNN/MobileNetV3-FPN for 10 epochs using input size 320×320 , batch size 1, AdamW with learning rate 10^{-3} and weight decay 10^{-4} , cosine learning-rate scheduling, GPU execution, and seeds $\{0, 1, 2\}$. For every seed, the checkpoint with the highest validation detection metric is selected and the test split is evaluated once. Seventh, the completed D07 semantic-segmentation rerun evaluates DT1D-Adapter, Linear probing, BitFit, and Full fine-tuning on Oxford-IIIT Pet with a ViT-B/16 dense decoder for 5 epochs using input size 224×224 , batch size 8, AdamW with learning rate 10^{-3} and weight decay 10^{-4} , cosine learning-rate scheduling, and seeds $\{0, 1, 2\}$. The best checkpoint for each seed is selected by validation mIoU and the test split is evaluated once. Eighth, C10 provides a complete three-seed Oxford-IIIT Pet/EfficientNet-B0 10-epoch comparison with validation-only method-level learning-rate selection, while C11 targets the corresponding 100-epoch setting but currently contains complete final results only for Linear probing and BitFit.

Table 2: Core ablation of DT1D-Adapter on DTD/ResNet-18 and Flowers102/ResNet-50. Results are reported as mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$. Test Acc@1 is evaluated at the best-validation checkpoint, and latency is reported per image.

Configuration	DTD / ResNet-18				Flowers102 / ResNet-50			
	Trainable params	Test Acc@1 (%)	Latency (ms/img)		Trainable params	Test Acc@1 (%)	Latency (ms/img)	
DT1D-Adapter	25,335	62.163 $\pm$ 0.406	1.758 $\pm$ 0.131		218,486	81.477 $\pm$ 0.551	6.177 $\pm$ 0.092	
Without weighted shift	25,319	61.986 $\pm$ 0.201	1.602 $\pm$ 0.041		218,454	81.417 $\pm$ 0.628	5.687 $\pm$ 0.038	
Without joint ℓ_1 projection	25,335	61.613 $\pm$ 0.354	1.074 $\pm$ 0.006		218,486	79.910 $\pm$ 0.316	6.114 $\pm$ 0.043	
Unrestricted axial depthwise 1D	89,399	61.418 $\pm$ 0.374	1.152 $\pm$ 0.004		722,550	80.577 $\pm$ 0.277	6.591 $\pm$ 0.074	

Ninth, C13 targets EuroSAT/MobileNetV3-Small for 25 epochs with batch size 32 and the same validation-only method-level learning-rate-selection rule; the supplied C13 archive currently contains complete final results only for Linear probing. Tenth, the completed D03 semantic-segmentation rerun evaluates DT1D-Adapter, Full fine-tuning, Linear probing, Conv-Adapter ($r = 4$), and BAM on Oxford-IIIT Pet using DeepLabV3-MobileNetV3 for 5 epochs with input size 224×224 , batch size 8, learning rate 10^{-3} , weight decay 10^{-4} , and seeds $\{0, 1, 2\}$. For each seed, the best checkpoint is selected by validation mIoU and the test split is evaluated once at that checkpoint.

For classification, top-1 accuracy is the primary metric. Where available, we also report top-5 accuracy, test loss, trainable parameters, parameter ratio, FP32 task storage, FLOPs, latency, throughput, training time, and memory use. For binary segmentation, we report foreground-class IoU and Dice together with global pixel accuracy, trainable parameters, task-specific FP32 storage, latency, training memory, and epoch time. For semantic segmentation, we report macro-class mIoU and Dice together with global pixel accuracy, parameter counts, task-specific FP32 storage, latency, throughput, inference and training memory, epoch time, and total training time. For object detection, we report AP₅₀, Recall₅₀, ground-truth and prediction counts, parameter counts, task-specific storage, latency, throughput, memory, epoch time, and total training time. Blank entries do not represent zero values; they indicate that the corresponding current rerun result has not yet been supplied.

5.2 Experimental results

5.2.1 Ablation and hyperparameter sensitivity

We evaluate four configurations to isolate the contributions of the structured kernel parameterization, the learned weighted shift, and the post-shift joint ℓ_1 projection: DT1D-Adapter, the structured core without the weighted shift, the structured core without the joint ℓ_1 projection, and an unrestricted axial depthwise 1D control.

The ablation results show that the weighted shift provides a modest accuracy increment over the no-shift structured core: +0.177 percentage point on DTD and +0.060 point on Flowers102. Removing the post-shift joint ℓ_1 projection produces a larger decrease, from 62.163% to 61.613% on DTD and from 81.477% to 79.910% on Flowers102. Relative to the unrestricted axial depthwise control, DT1D-Adapter reduces trainable parameters from 89,399 to 25,335 on DTD and from 722,550 to

Table 3: Current three-seed comparison on DTD with an ImageNet-pretrained ResNet-50 backbone trained for 100 epochs using batch size 128. Values are populated only for completed rows in the supplied current result set; all other cells are intentionally left blank.

(a) Predictive performance and parameter/storage efficiency								
Method	Trainable params	Total params	Trainable (%)	FP32 storage/task (MiB)	Best val Acc@1 (%)	Test Acc@1 (%)	Test Acc@5 (%)	Test loss
DT1D-Adapter								
Full fine-tuning								
Linear probing	96,303	23,604,335	0.408		0.367 71.028 $\pm$ 0.081	70.691 $\pm$ 0.349	92.748 $\pm$ 0.163	1.120 $\pm$ 0.049
BitFit [14]	122,863	23,604,335	0.521		0.469 71.418 $\pm$ 0.494	72.163 $\pm$ 0.342	92.695 $\pm$ 0.630	1.403 $\pm$ 0.097
SSF [16]								
Conv-Adapter [36] ($r = 2$)								
Conv-Adapter [36] ($r = 4$)								
Conv-Adapter [36] ($r = 6$)	6,810,724	30,318,756	22.464		25.981 70.390 $\pm$ 0.111	71.294 $\pm$ 0.111	92.411 $\pm$ 0.342	1.689 $\pm$ 0.317
Conv-Adapter [36] ($r = 8$)	5,143,183	28,651,215	17.951		19.620 70.177 $\pm$ 0.494	71.011 $\pm$ 0.435	92.376 $\pm$ 0.187	1.666 $\pm$ 0.040
Residual Adapters [38]								
(b) Computational and runtime efficiency								
Method	FLOPs (G)	Latency (ms/img)	FPS	Training time (s)	Peak train memory (MB)	Peak inference memory (MB)		
DT1D-Adapter								
Full fine-tuning								
Linear probing	4.110	1.425 $\pm$ 0.010	701.609 $\pm$ 4.904	2393.565 $\pm$ 15.112	868.530	194.592		
BitFit [14]	4.110	1.473 $\pm$ 0.027	678.937 $\pm$ 12.684	2470.389 $\pm$ 5.889	5603.667	194.592		
SSF [16]								
Conv-Adapter [36] ($r = 2$)								
Conv-Adapter [36] ($r = 4$)								
Conv-Adapter [36] ($r = 6$)	5.208	2.165 $\pm$ 0.022	462.017 $\pm$ 4.653	2622.093 $\pm$ 36.883	6275.117	238.792		
Conv-Adapter [36] ($r = 8$)	4.938	1.962 $\pm$ 0.035	509.854 $\pm$ 9.001	2608.979 $\pm$ 30.208	6149.434	235.568		
Residual Adapters [38]								

Blank cells indicate method results not yet available in the current result set.

218,486 on Flowers102 while slightly increasing mean test Acc@1 in both settings. These results support a parameter-compact structured operating point; they do not establish universal wall-clock superiority.

5.2.2 CNN-backbone image classification

The current DTD/ResNet-50 and Flowers102/ResNet-50 reruns are only partially complete. Tables 3 and 4 therefore populate only completed method rows from the supplied current result archives and retain the remaining rows as blanks.

The supplied C03 archive completes the Flowers102/ResNet-18 10-epoch, batch-size-64 comparison in Table 5. It contains successful three-seed runs for DT1D-Adapter, Full fine-tuning, Linear probing, BitFit, SSF, and Conv-Adapter with bottleneck ratios $r \in \{2, 4, 6, 8\}$ under the same validation-only learning-rate selection protocol. Methods not present in C03 remain blank rather than carrying forward legacy values.

The completed C03 comparison shows that Full fine-tuning obtains the highest mean test Acc@1 at $88.415 \pm 0.206\%$. Among the supplied parameter-efficient methods, BitFit ($83.157 \pm 0.474\%$), SSF ($83.103 \pm 0.818\%$), and DT1D-Adapter ($83.011 \pm 0.244\%$) form the strongest accuracy group. DT1D-Adapter uses 53,550 trainable parameters (0.477% of the 11.230M-parameter model), close to Linear probing, BitFit, and SSF, but its functional axial operations raise the profiler estimate to 3.686 G FLOPs

Table 4: Current three-seed comparison on Flowers102 with an ImageNet-pretrained ResNet-50 backbone trained for 100 epochs using batch size 128. Values are populated only for completed rows in the supplied current result set; all other cells are intentionally left blank.

(a) Predictive performance and parameter/storage efficiency								
Method	Trainable params	Total params	Trainable (%)	FP32 storage/task (MiB)	Best val Acc@1 (%)	Test Acc@1 (%)	Test Acc@5 (%)	Test loss
DT1D-Adapter								
Full fine-tuning								
Linear probing	208,998	23,717,030	0.881		0.797 84.118 $\pm$ 0.170	82.431 $\pm$ 0.123	94.102 $\pm$ 0.146	0.756 $\pm$ 0.011
BitFit [14]	235,558	23,717,030	0.993		0.899 85.882 $\pm$ 0.784	84.903 $\pm$ 0.593	95.972 $\pm$ 0.122	0.606 $\pm$ 0.010
SSF [16]								
Conv-Adapter [36] ($r = 2$)								
Conv-Adapter [36] ($r = 4$)								
Conv-Adapter [36] ($r = 6$)								
Conv-Adapter [36] ($r = 8$)								
Residual Adapters [38]								
(b) Computational and runtime efficiency								
Method	FLOPs (G)	Latency (ms/img)	FPS	Training time (s)	Peak train memory (MB)	Peak inference memory (MB)		
DT1D-Adapter								
Full fine-tuning								
Linear probing	4.110	1.391 $\pm$ 0.006	719.067 $\pm$ 3.063	1513.376 $\pm$ 20.313	869.833	195.022		
BitFit [14]	4.110	1.420 $\pm$ 0.003	704.100 $\pm$ 1.349	1545.136 $\pm$ 15.266	5605.399	195.022		
SSF [16]								
Conv-Adapter [36] ($r = 2$)								
Conv-Adapter [36] ($r = 4$)								
Conv-Adapter [36] ($r = 6$)								
Conv-Adapter [36] ($r = 8$)								
Residual Adapters [38]								

Blank cells indicate method results not yet available in the current result set.

and 1.770 ± 0.117 ms/image latency. The C03 result therefore supports parameter compactness without implying a wall-clock advantage over the simpler PEFT baselines.

The completed C10 rerun shows a narrow accuracy spread among the PEFT baselines on Oxford-IIIT Pet/EfficientNet-B0 at 10 epochs. Conv-Adapter ($r = 4$) records the highest supplied mean test Acc@1 at $91.987 \pm 0.179\%$, while DT1D-Adapter reaches $91.905 \pm 0.109\%$ with 48,595 trainable parameters, corresponding to 1.198% of the 4.056M-parameter model and approximately 0.185 MiB of FP32 task-specific storage. Linear probing and SSF both reach 91.878% mean test Acc@1, and the supplied Full fine-tuning run reaches $89.552 \pm 0.659\%$. The runtime profile makes the trade-off explicit: DT1D-Adapter requires 0.858 G FLOPs and 3.995 ± 0.466 ms/image (252.500 ± 27.868 FPS), whereas Linear probing requires 0.400 G FLOPs and 0.895 ± 0.209 ms/image and Conv-Adapter ($r = 4$) requires 0.418 G FLOPs and 1.242 ± 0.157 ms/image. Thus, the C10 result supports parameter/storage compactness but not a claim of lower inference latency. The C11 100-epoch archive is incomplete; among the two methods with complete three-seed final results, Linear probing reaches $91.678 \pm 0.265\%$ with 0.895 ± 0.114 ms/image latency, while BitFit reaches $91.251 \pm 0.330\%$ with 0.956 ± 0.059 ms/image latency. No conclusion is drawn for the missing C11 methods until their final runs are supplied.

The current C12 results place the completed Caltech101/ResNet-18 methods in a narrow accuracy range. BitFit has the highest completed mean test Acc@1 at $94.887 \pm$

Table 5: Three-seed accuracy, parameter/storage, and runtime comparison on Flowers102 with an ImageNet-pretrained ResNet-18 backbone trained for 10 epochs using batch size 64. Results are reported over independent seeds $\{0, 1, 2\}$. For each seed, the test set is evaluated at the checkpoint with the highest validation Acc_1 . Runtime profiling uses 224×224 inputs with profile batch size 16 on an NVIDIA Tesla T4.

(a) Predictive performance and parameter/storage efficiency

Method	Trainable params	Total params (M)	Trainable (%)	FP32 storage/task (MB)	Best val Acc_1 (%)	Test Acc_1 at best val (%)
DT1D-Adapter	53,550	11.230	0.477	0.214	84.542 ± 0.786	83.011 ± 0.244
Full fine-tuning	11,228,838	11.229	100.000	44.915	91.765 ± 0.259	88.415 ± 0.206
Linear probing	52,326	11.229	0.466	0.209	81.961 ± 0.294	80.490 ± 0.461
BitFit [14]	57,126	11.229	0.509	0.229	85.294 ± 0.995	83.157 ± 0.474
SSF [16]	56,166	11.233	0.500	0.225	83.922 ± 0.980	83.103 ± 0.818
BAM [42]						
Residual Adapters [38]						
LoRA-Conv [13]						
Side-Tuning [69]						
Conv-Adapter [36] ($r = 8$)	228,566	11.405	2.004	0.914	84.118 ± 1.968	81.899 ± 0.685
Conv-Adapter [36] ($r = 6$)	285,570	11.462	2.491	1.142	82.614 ± 2.206	80.322 ± 1.690
Conv-Adapter [36] ($r = 4$)	404,806	11.581	3.495	1.619	81.961 ± 1.444	79.970 ± 0.939
Conv-Adapter [36] ($r = 2$)	757,286	11.934	6.346	3.029	82.320 ± 1.374	80.485 ± 0.918

(b) Computational, runtime, and memory efficiency

Method	FLOPs (G)	Latency (ms/img)	FPS	Peak train memory (MB)	Peak inference memory (MB)	Train time (s)
DT1D-Adapter	3.686	1.770 ± 0.117	566.421 ± 36.216	897.613	127.926	138.071 ± 0.138
Full fine-tuning	1.819	0.546 ± 0.007	1831.536 ± 24.530	959.242	120.960	88.878 ± 1.116
Linear probing	1.819	0.548 ± 0.007	1825.648 ± 23.080	336.219	120.960	131.959 ± 1.159
BitFit [14]	1.819	0.545 ± 0.002	1834.391 ± 6.970	851.074	120.960	132.978 ± 0.562
SSF [16]	1.819	0.599 ± 0.008	1668.718 ± 23.610	600.114	120.976	132.597 ± 0.275
BAM [42]						
Residual Adapters [38]						
LoRA-Conv [13]						
Side-Tuning [69]						
Conv-Adapter [36] ($r = 8$)	1.845	0.627 ± 0.002	1593.857 ± 4.622	570.274	121.635	132.149 ± 2.149
Conv-Adapter [36] ($r = 6$)	1.853	0.625 ± 0.002	1599.951 ± 4.988	578.688	121.851	136.481 ± 2.630
Conv-Adapter [36] ($r = 4$)	1.872	0.633 ± 0.005	1579.792 ± 12.870	595.580	122.307	135.171 ± 1.453
Conv-Adapter [36] ($r = 2$)	1.925	0.640 ± 0.015	1563.334 ± 37.210	644.254	123.650	140.239 ± 1.238

FP32 storage/task is computed as four bytes per trainable task-specific parameter. Latency, FPS, and training time are reported as mean $\pm$ sample standard deviation over the three seeds; parameter counts, FLOPs, and the reported peak-memory values are constant across the three runs for each method in this experiment. The FLOP values follow the repository profiler. For DT1D-Adapter, the profiler explicitly adds the functional axial operations that are not represented as ordinary traced modules; these figures should therefore be interpreted as implementation-level compute estimates rather than exact hardware-independent operation counts.

Blank cells indicate methods not included in the supplied C03 archive.

0.240%, followed by Conv-Adapter at $94.771 \pm 1.072\%$, BAM at $94.617 \pm 0.371\%$, and DT1D-Adapter at $94.387 \pm 0.569\%$. DT1D-Adapter uses 53,037 trainable parameters, close to Linear probing, SSF, and BitFit; its C12 profile reports 3.686 G FLOPs and 2.066 ± 0.407 ms/image latency. Full fine-tuning, LoRA-Conv, and Side-Tuning remain blank because those current C12 runs are not present in the supplied archive.

Table 6: Three-seed accuracy, parameter, and storage results on Oxford-IIIT Pet with EfficientNet-B0 trained for 10 epochs using batch size 64. A single learning rate is selected for each method by mean best-validation Acc@1 across seeds $\{0, 1, 2\}$ from the common grid $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 5 \times 10^{-3}\}$, and each seed is evaluated once on the test set at its best-validation checkpoint. Values that vary across seeds are reported as mean $\pm$ sample standard deviation.

Method	Trainable params	Total params	Trainable (%)	FP32 storage/task (MiB)	Best val Acc@1 (%)	Test Acc@1 (%)	Test Acc@5 (%)	Test loss	
DT1D-Adapter	48,595	4,056,143	1.198		0.185	92.618 ± 1.699	91.905 ± 0.109	99.700 ± 0.072	0.271 ± 0.006
Full fine-tuning	4,054,945	4,054,945	100.000		15.468	91.531 ± 0.905	89.552 ± 0.659	99.255 ± 0.031	0.340 ± 0.016
Linear probing	47,397	4,054,945	1.169		0.181	92.889 ± 1.340	91.878 ± 0.191	99.646 ± 0.072	0.255 ± 0.004
Conv-Adapter [36] ($r = 4$)	207,113	4,214,661	4.914		0.790	93.025 ± 1.660	91.987 ± 0.179	99.646 ± 0.055	0.280 ± 0.009
Conv-Adapter [36] ($r = 2$)	366,829	4,374,377	8.386		1.399	93.161 ± 1.218	91.651 ± 0.437	99.628 ± 0.042	0.282 ± 0.014
Conv-Adapter [36] ($r = 6$)	152,877	4,160,425	3.675		0.583	93.207 ± 1.338	91.769 ± 0.428	99.582 ± 0.083	0.285 ± 0.005
Conv-Adapter [36] ($r = 8$)	127,255	4,134,803	3.078		0.485	93.116 ± 1.139	91.823 ± 0.361	99.555 ± 0.042	0.280 ± 0.013
SSF [16]	51,013	4,058,561	1.257		0.195	92.889 ± 1.225	91.878 ± 0.233	99.618 ± 0.119	0.249 ± 0.008
BitFit [14]	77,745	4,054,945	1.917		0.297	93.207 ± 1.184	91.842 ± 0.873	99.582 ± 0.063	0.267 ± 0.012
Prompt/VPT [15]									

Blank cells indicate results not available in the supplied C10 archive. FP32 storage/task is computed from the trainable-parameter count using four bytes per parameter.

Table 7: Three-seed efficiency and runtime results for the C10 Oxford-IIIT Pet/EfficientNet-B0 10-epoch experiment. FLOPs and profiler inference memory are architecture-level quantities; latency, throughput, training time, and epoch time are reported as mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$. Profiling uses 224×224 inputs and profile batch size 16.

Method	FLOPs (G)	Latency (ms/img)	FPS	Train time (s)	Epoch time (s)	Peak train memory (MB)	Peak inference memory (MB)
DT1D-Adapter	0.858	3.995 $\pm$ 0.466	252.500 $\pm$ 27.868	254.893 $\pm$ 2.535	20.282 $\pm$ 0.077	2715.984	113.891
Full fine-tuning	0.400	0.748 $\pm$ 0.001	1337.496 $\pm$ 2.443	168.802 $\pm$ 17.358	13.373 $\pm$ 1.355	2847.094	113.651
Linear probing	0.400	0.895 $\pm$ 0.209	1153.934 $\pm$ 239.490	219.959 $\pm$ 0.371	17.092 $\pm$ 0.124	390.575	113.643
Conv-Adapter [36] ($r = 4$)	0.418	1.242 $\pm$ 0.157	814.260 $\pm$ 111.021	231.187 $\pm$ 0.999	18.052 $\pm$ 0.218	2537.708	114.264
Conv-Adapter [36] ($r = 2$)	0.436	1.375 $\pm$ 0.222	740.934 $\pm$ 125.287	233.791 $\pm$ 12.732	18.318 $\pm$ 0.971	2577.825	114.868
Conv-Adapter [36] ($r = 6$)	0.412	1.580 $\pm$ 0.078	634.087 $\pm$ 30.575	232.263 $\pm$ 2.371	18.125 $\pm$ 0.261	2527.269	114.056
Conv-Adapter [36] ($r = 8$)	0.409	1.386 $\pm$ 0.260	738.858 $\pm$ 141.259	230.261 $\pm$ 2.460	18.112 $\pm$ 0.283	2523.712	113.958
SSF [16]	0.400	1.024 $\pm$ 0.171	996.028 $\pm$ 173.789	243.797 $\pm$ 3.266	19.227 $\pm$ 0.263	2504.972	119.789
BitFit [14]	0.400	0.891 $\pm$ 0.128	1138.954 $\pm$ 168.141	227.078 $\pm$ 2.338	17.831 $\pm$ 0.298	2802.236	113.644
Prompt/VPT [15]							

Blank cells indicate results not available in the supplied C10 archive. Latency, FPS, FLOPs, and peak inference memory are taken from the released efficiency profiler; training-memory and timing values come from the completed training runs.

Table 8: Current three-seed accuracy, parameter, and storage results on Oxford-IIIT Pet with EfficientNet-B0 trained for 100 epochs using batch size 64. The supplied C11 archive is partial: only Linear probing and BitFit contain complete final results for seeds $\{0, 1, 2\}$. Values that vary across seeds are reported as mean $\pm$ sample standard deviation; incomplete method rows remain blank.

Method	Trainable params	Total params	Trainable (%)	FP32 storage/task (MiB)	Best val Acc@1 (%)	Test Acc@1 (%)	Test Acc@5 (%)	Test loss
DT1D-Adapter								
Full fine-tuning								
Linear probing	47,397	4,054,945	1.169		0.181 93.207 $\pm$ 1.078	91.678 $\pm$ 0.265	99.509 $\pm$ 0.072	0.281 $\pm$ 0.021
Conv-Adapter [36] ($r = 4$)								
Conv-Adapter [36] ($r = 2$)								
Conv-Adapter [36] ($r = 6$)								
Conv-Adapter [36] ($r = 8$)								
SSF [16]								
BitFit [14]	77,745	4,054,945	1.917		0.297 93.659 $\pm$ 1.621	91.251 $\pm$ 0.330	99.382 $\pm$ 0.063	0.377 $\pm$ 0.033
Prompt/VPT [15]								

Blank cells indicate method results not complete in the supplied C11 archive. FP32 storage/task is computed from the trainable-parameter count using four bytes per parameter.

Table 9: Current efficiency and runtime results for the C11 Oxford-IIIT Pet/EfficientNet-B0 100-epoch experiment. Only Linear probing and BitFit have complete three-seed final results in the supplied archive. Profiling uses 224×224 inputs and profile batch size 16.

Method	FLOPs (G)	Latency (ms/img)	FPS	Train time (s)	Epoch time (s)	Peak train memory (MB)	Peak inference memory (MB)
DT1D-Adapter							
Full fine-tuning							
Linear probing	0.400	0.895 $\pm$ 0.114	1128.917 $\pm$ 143.909	2104.642 $\pm$ 20.456	16.615 $\pm$ 0.204	390.575	113.643
Conv-Adapter [36] ($r = 4$)							
Conv-Adapter [36] ($r = 2$)							
Conv-Adapter [36] ($r = 6$)							
Conv-Adapter [36] ($r = 8$)							
SSF [16]							
BitFit [14]	0.400	0.956 $\pm$ 0.059	1048.913 $\pm$ 66.429	2139.431 $\pm$ 19.836	17.017 $\pm$ 0.234	2802.236	113.644
Prompt/VPT [15]							

Blank cells indicate method results not complete in the supplied C11 archive. Latency, FPS, FLOPs, and peak inference memory are taken from the released efficiency profiler; training-memory and timing values come from the completed training runs.

Table 10: Accuracy and parameter results on Caltech101 with a ResNet-18 backbone trained for 10 epochs using batch size 64. Results are mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$; for each seed, the test set is evaluated at the checkpoint with the highest validation Acc₁. Trainable and total parameter counts are fixed by the corresponding method configuration.

Method	Trainable params	Total params	Best val acc. (%)	Best epoch	Test Acc ₁ (%)	Test Acc ₅ (%)	Test loss
DT1D-Adapter	53,037	11,229,549	94.854 $\pm$ 0.580	7.667 $\pm$ 0.577	94.387 $\pm$ 0.569	99.116 $\pm$ 0.266	0.228 $\pm$ 0.006
Full fine-tuning							
Linear probing	51,813	11,228,325	93.894 $\pm$ 0.610	6.000 $\pm$ 1.732	93.349 $\pm$ 0.656	98.962 $\pm$ 0.115	0.255 $\pm$ 0.022
Conv-Adapter [36] ($r = 4$)	404,293	11,580,805	94.585 $\pm$ 0.115	7.000 $\pm$ 1.732	94.771 $\pm$ 1.072	99.154 $\pm$ 0.067	0.229 $\pm$ 0.021
BAM [42]	129,397	11,306,149	94.547 $\pm$ 0.544	7.667 $\pm$ 1.528	94.617 $\pm$ 0.371	99.308 $\pm$ 0.115	0.196 $\pm$ 0.018
Residual Adapters [38]	138,861	11,315,613	94.163 $\pm$ 0.333	5.667 $\pm$ 1.155	94.233 $\pm$ 1.112	99.000 $\pm$ 0.290	0.249 $\pm$ 0.058
SSF [16]	55,653	11,232,165	94.278 $\pm$ 0.655	7.000 $\pm$ 1.000	94.002 $\pm$ 1.006	99.193 $\pm$ 0.346	0.238 $\pm$ 0.043
LoRA-Conv [13]							
BitFit [14]	56,613	11,228,325	95.315 $\pm$ 0.133	7.333 $\pm$ 1.528	94.887 $\pm$ 0.240	99.116 $\pm$ 0.290	0.208 $\pm$ 0.027
Side-Tuning [69]							

Blank cells indicate current C12 results not yet available for Full fine-tuning, LoRA-Conv, and Side-Tuning.

Table 11: Efficiency and runtime results on Caltech101 with a ResNet-18 backbone trained for 10 epochs using batch size 64. Latency, FPS, training time, and epoch time are reported as mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$; FLOPs are architecture-level quantities and are unchanged across seeds.

Method	FLOPs (G)	Latency (ms/img)	FPS	Train time (s)	Epoch time (s)
DT1D-Adapter	3.686	2.066 ± 0.407	498.651 ± 110.640	368.394 ± 1.492	33.287 ± 0.113
Full fine-tuning					
Linear probing	1.819	0.504 ± 0.003	1983.352 ± 13.669	252.866 ± 4.450	22.020 ± 0.357
Conv-Adapter [36] ($r = 4$)	1.872	0.612 ± 0.004	1633.496 ± 10.273	174.410 ± 0.911	15.239 ± 0.100
BAM [42]	1.824	0.575 ± 0.002	1738.976 ± 4.715	272.247 ± 3.909	23.813 ± 0.402
Residual Adapters [38]	1.832	0.607 ± 0.002	1646.252 ± 6.711	169.362 ± 1.249	14.861 ± 0.111
SSF [16]	1.819	0.581 ± 0.013	1722.256 ± 37.576	287.861 ± 22.735	25.376 ± 2.028
LoRA-Conv [13]					
BitFit [14]	1.819	0.498 ± 0.003	2006.915 ± 11.521	253.714 ± 1.699	22.312 ± 0.175
Side-Tuning [69]					

Blank cells indicate current C12 results not yet available for Full fine-tuning, LoRA-Conv, and Side-Tuning.

The additional CNN-only evaluation contains exactly four small-dataset experiments rather than a Cartesian product of datasets and backbones. N01 uses EfficientNet-B0 on STL10 for 10 epochs; N02 uses MobileNetV3-Small on Oxford Flowers17 for 10 epochs; N03 uses DenseNet-121 [9] on STL10 for 100 epochs; and N04 uses ResNet-18 on Oxford Flowers17 for 100 epochs. This provides compound-scaled, mobile-oriented, densely connected, and residual backbone coverage while keeping the additional experimental scope compact. The same ten-method baseline family, seeds $\{0, 1, 2\}$, and validation-only shared-learning-rate rule are used in all four settings.

Table 12: Reserved v10 result-entry template for N01: STL10 [74] with EfficientNet-B0, trained for 10 epochs using batch size 64 and seeds $\{0, 1, 2\}$. The table is intentionally blank until each method has a complete three-seed aggregate after validation-only shared-learning-rate selection.

Method	Trainable params	Trainable (%)	Best val Acc@1 (%)	Test Acc@1 (%)	FLOPs (G)	Latency (ms/img)
DT1D-Adapter						
Full fine-tuning						
Linear probing						
BitFit [14]						
SSF [16]						
Conv-Adapter [36] ($r = 4$)						
BAM [42]						
Residual Adapters [38]						
LoRA-Conv [13]						
Side-Tuning [69]						

Blank cells are deliberate. Fill a method row only from the complete three-seed aggregate for this exact dataset/backbone/epoch setting; do not insert partial-session values.

Table 13: Reserved v10 result-entry template for N02: Oxford Flowers17 [75] with MobileNetV3-Small, trained for 10 epochs using batch size 64 and seeds $\{0, 1, 2\}$. The table is intentionally blank until each method has a complete three-seed aggregate after validation-only shared-learning-rate selection.

Method	Trainable params	Trainable (%)	Best val Acc@1 (%)	Test Acc@1 (%)	FLOPs (G)	Latency (ms/img)
DT1D-Adapter						
Full fine-tuning						
Linear probing						
BitFit [14]						
SSF [16]						
Conv-Adapter [36] ($r = 4$)						
BAM [42]						
Residual Adapters [38]						
LoRA-Conv [13]						
Side-Tuning [69]						

Blank cells are deliberate. Fill a method row only from the complete three-seed aggregate for this exact dataset/backbone/epoch setting; do not insert partial-session values.

Table 14: Reserved v10 result-entry template for N03: STL10 [74] with DenseNet-121 [9], trained for 100 epochs using batch size 64 and seeds {0, 1, 2}. The table is intentionally blank until each method has a complete three-seed aggregate after validation-only shared-learning-rate selection.

Method	Trainable params	Trainable (%)	Best val Acc@1 (%)	Test Acc@1 (%)	FLOPs (G)	Latency (ms/img)
DT1D-Adapter						
Full fine-tuning						
Linear probing						
BitFit [14]						
SSF [16]						
Conv-Adapter [36] ($r = 4$)						
BAM [42]						
Residual Adapters [38]						
LoRA-Conv [13]						
Side-Tuning [69]						
<i>Blank cells are deliberate. Fill a method row only from the complete three-seed aggregate for this exact dataset/backbone/epoch setting; do not insert partial-session values.</i>						

Table 15: Reserved v10 result-entry template for N04: Oxford Flowers17 [75] with ResNet-18, trained for 100 epochs using batch size 64 and seeds {0, 1, 2}. The table is intentionally blank until each method has a complete three-seed aggregate after validation-only shared-learning-rate selection.

Method	Trainable params	Trainable (%)	Best val Acc@1 (%)	Test Acc@1 (%)	FLOPs (G)	Latency (ms/img)
DT1D-Adapter						
Full fine-tuning						
Linear probing						
BitFit [14]						
SSF [16]						
Conv-Adapter [36] ($r = 4$)						
BAM [42]						
Residual Adapters [38]						
LoRA-Conv [13]						
Side-Tuning [69]						
<i>Blank cells are deliberate. Fill a method row only from the complete three-seed aggregate for this exact dataset/backbone/epoch setting; do not insert partial-session values.</i>						

5.2.3 Transformer-backbone image classification

The supplied current ViT result set contains complete three-seed results for Flowers102 at batch sizes 16 and 32 and for VTAB-Caltech101, VTAB-DTD, and VTAB-EuroSAT at batch size 32.

For Flowers102, DT1D-Adapter obtains $96.092 \pm 0.203\%$ test Acc@1 at batch size 32 and $95.864 \pm 0.191\%$ at batch size 16, in both cases with 84,234 trainable parameters. At batch size 32, Pfeiffer Adapter records the highest mean test Acc@1

Table 16: Three-seed ViT-B/16 [11] results on Flowers102. Models are trained for 10 epochs at 224×224 resolution using seeds $\{0, 1, 2\}$. The supplied current result set contains complete batch-size-16 and batch-size-32 three-seed results.

BS	Method	Opt. / LR	Trainable params	Trainable (%)	FP32 storage (MiB)	Test Acc@1 (%)	Test Acc@5 (%)	Test loss	Test batch time (s)	Peak GPU mem. (GB)
32	DT1D-Adapter	AdamW / 0.01	84,234	0.098	0.321	96.092 $\pm$ 0.203				
	Full fine-tuning	AdamW / 2.5×10^{-4}	85,877,094	100.000	327.595	95.858 $\pm$ 0.156				
	Linear probing	SGD / 0.0625	78,438	0.091	0.299	95.815 $\pm$ 0.377				
	VPT [15]	SGD / 0.125	82,278	0.096	0.314	95.858 $\pm$ 0.272				
	Pfeiffer Adapter	AdamW / 2.5×10^{-3}	972,966	1.121	3.712	96.867 $\pm$ 0.047				
16	DT1D-Adapter	AdamW / 0.01	84,234	0.098	0.321	95.864 $\pm$ 0.191				
	Full fine-tuning	AdamW / 3.125×10^{-4}	85,877,094	100.000	327.595	95.511 $\pm$ 0.658				
	Linear probing	SGD / 0.03125	78,438	0.091	0.299	95.533 $\pm$ 0.292				
	VPT [15]	SGD / 0.625	82,278	0.096	0.314	95.072 $\pm$ 0.703				
	Pfeiffer Adapter	AdamW / 2.5×10^{-3}	972,966	1.121	3.712	96.471 $\pm$ 0.181				

Blank ancillary cells indicate values not present in the supplied current result set.

Table 17: Three-seed comparison on VTAB-Caltech101 using ViT-B/16 [11]. The VTAB-1k protocol uses **train800** for training, **val200** for validation, and the official test set for testing. All methods are trained for 10 epochs with batch size 32 and 224×224 inputs. Results are mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$.

Method	Opt. / LR	Trainable params	Trainable (%)	FP32 storage (MiB)	Best val Acc@1 (%)	Test Acc@1 (%)
DT1D-Adapter	AdamW / 0.01	84,234	0.098	0.321	82.167 $\pm$ 1.155	82.249 $\pm$ 1.078
Full fine-tuning	AdamW / 2.5×10^{-4}	85,877,094	100.000	327.595	82.667 $\pm$ 1.607	72.770 $\pm$ 4.338
Linear probing	SGD / 0.0625	78,438	0.091	0.299	82.833 $\pm$ 2.021	80.736 $\pm$ 2.133
VPT [15]	SGD / 0.125	86,118	0.100	0.329	83.000 $\pm$ 1.803	82.495 $\pm$ 1.335
Pfeiffer Adapter	AdamW / 0.005	972,966	1.121	3.712	82.000 $\pm$ 2.784	75.920 $\pm$ 0.740

Table 18: Three-seed comparison on VTAB-DTD using ViT-B/16 [11]. The VTAB-1k protocol uses **train800** for training, **val200** for validation, and the official test set for testing. All methods are trained for 10 epochs with batch size 32 and 224×224 inputs. Results are mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$.

Method	Opt. / LR	Trainable params	Trainable (%)	FP32 storage (MiB)	Best val Acc@1 (%)	Test Acc@1 (%)
DT1D-Adapter	AdamW / 0.01	41,939	0.049	0.160	48.667 $\pm$ 4.041	49.450 $\pm$ 1.413
Full fine-tuning	AdamW / 6.25×10^{-4}	85,834,799	100.000	327.434	52.833 $\pm$ 0.764	53.280 $\pm$ 0.826
Linear probing	SGD / 6.25	36,143	0.042	0.138	57.333 $\pm$ 0.764	55.106 $\pm$ 0.555
VPT [15]	SGD / 1.25	43,823	0.051	0.167	56.167 $\pm$ 2.021	55.142 $\pm$ 1.518
Pfeiffer Adapter	AdamW / 0.01	930,671	1.073	3.550	50.833 $\pm$ 3.253	51.223 $\pm$ 1.778

Blank cells indicate results not yet available in the current result set.

at $96.867 \pm 0.047\%$ but uses 972,966 trainable parameters, whereas DT1D-Adapter exceeds Full fine-tuning ($95.858 \pm 0.156\%$), VPT ($95.858 \pm 0.272\%$), and Linear probing ($95.815 \pm 0.377\%$) in mean test Acc@1. At batch size 16, Pfeiffer Adapter similarly records the highest mean at $96.471 \pm 0.181\%$. On VTAB-Caltech101, DT1D-Adapter reaches $82.249 \pm 1.078\%$ with 84,234 trainable parameters, closely matching VPT at $82.495 \pm 1.335\%$ with 86,118 trainable parameters and exceeding Linear probing, Pfeiffer Adapter, and Full fine-tuning in mean test Acc@1. On VTAB-DTD, DT1D-Adapter reaches $49.450 \pm 1.413\%$ with 41,939 trainable parameters, while VPT and

Table 19: Three-seed comparison on VTAB-EuroSAT using ViT-B/16 [11]. The VTAB-1k protocol uses **train800** for training, **val200** for validation, and the official test set for testing. All methods are trained for 10 epochs with batch size 32 and 224×224 inputs. Results are mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$.

Method	Opt. / LR	Trainable params	Trainable (%)	FP32 storage (MiB)	Best val Acc@1 (%)	Test Acc@1 (%)
DT1D-Adapter	AdamW / 0.01	13,486	0.016	0.051	78.667 ± 4.041	77.179 ± 0.593
Full fine-tuning	AdamW / 2.5×10^{-4}	85,806,346	100.000	327.325	94.833 ± 1.528	93.549 ± 0.224
Linear probing	SGD / 6.25	7,690	0.009	0.029	89.500 ± 1.803	87.432 ± 0.750
VPT [15]	SGD / 3.125	15,370	0.018	0.059	87.000 ± 2.784	87.222 ± 2.135
Pfeiffer Adapter	AdamW / 0.01	902,218	1.041	3.442	93.333 ± 1.443	92.994 ± 0.391

Blank cells indicate results not yet available in the current result set.

Linear probing reach $55.142 \pm 1.518\%$ and $55.106 \pm 0.555\%$, respectively. On VTAB-EuroSAT, DT1D-Adapter reaches $77.179 \pm 0.593\%$ with 13,486 trainable parameters, while Full fine-tuning and Pfeiffer Adapter reach $93.549 \pm 0.224\%$ and $92.994 \pm 0.391\%$, respectively. These completed current ViT results therefore support a compact spatial adaptation interpretation but also show clear limitation cases in which parameter compactness does not translate into the strongest raw accuracy.

5.2.4 Efficiency profiling

The supplied C13 archive corresponds to the standalone EuroSAT/MobileNetV3-Small efficiency tables below. C13 is currently partial: only Linear probing contains complete three-seed final results, so that row is populated and all incomplete method rows remain blank.

Table 20: Current EuroSAT accuracy, parameter, storage, validation, and FLOPs results with MobileNetV3-Small. The supplied C13 archive currently contains complete final results only for Linear probing over seeds $\{0, 1, 2\}$. Values that vary across seeds are reported as mean $\pm$ sample standard deviation. Profiling uses input size 224×224 and profile batch size 16; training uses batch size 32.

Method	Trainable params	Total params	Trainable (%)	FP32 storage/task (MiB)	Best val Acc@1 (%)	Test Acc@1 (%)	Test Acc@5 (%)	Test loss	FLOPs (G)
DT1D-Adapter									
Full fine-tuning									
Linear probing	10,250	1,528,106	0.671	0.039	96.556 ± 0.556	95.901 ± 0.429	99.963 ± 0.000	0.134 ± 0.014	0.059
Conv-Adapter [36] ($r = 4$)									
BAM [42]									

Blank cells indicate method results not complete in the supplied C13 archive. FP32 storage/task is computed from the trainable-parameter count using four bytes per parameter.

For the one complete C13 method, Linear probing reaches $95.901 \pm 0.429\%$ test Acc@1 and $99.963 \pm 0.000\%$ test Acc@5 with 10,250 trainable parameters (0.671% of 1,528,106 total parameters) and approximately 0.039 MiB of FP32 task-specific storage. Its profiler result is 0.059 G FLOPs, 0.758 ± 0.010 ms/image latency, and 1319.887 ± 17.330 FPS, with 39.560 MB peak inference memory. The remaining C13 rows are intentionally left blank until complete three-seed results are available.

Table 21: Current EuroSAT latency, throughput, memory, and training-time results with MobileNetV3-Small. The supplied C13 archive currently contains complete final results only for Linear probing over seeds $\{0, 1, 2\}$. Values that vary across seeds are reported as mean $\pm$ sample standard deviation.

Method	Latency (ms/img)	FPS	Peak Infer Mem. (MB)	Peak Train Mem. (MB)	Epoch Time (s)	Train Time (s)
DT1D-Adapter						
Full fine-tuning						
Linear probing	3.758 $\pm$ 0.010	1319.887 $\pm$ 17.330	39.560	72.486	53.362 $\pm$ 1.714	1494.007 $\pm$ 44.138
Conv-Adapter [36]						
($r = 4$)						
BAM [42]						

Blank cells indicate method results not complete in the supplied C13 archive. Latency, FPS, and peak inference memory are taken from the released efficiency profiler; training-memory and timing values come from the completed training runs.

5.2.5 Binary mask segmentation

The completed D05A/D05B reruns provide three-seed PennFudan binary-mask segmentation results for DeepLabV3-MobileNetV3-Large, while the completed D04A/D04B reruns provide three-seed PennFudan results for the ViT-B/16 dense decoder.

The PennFudan results show a clear separation between unrestricted full fine-tuning and the PEFT methods. Full fine-tuning obtains the highest mean IoU at $67.853 \pm 1.604\%$. Among the supplied PEFT methods, LoRA-Conv gives the strongest segmentation accuracy with $49.860 \pm 1.334\%$ IoU and $66.535 \pm 1.184\%$ Dice. DT1D-Adapter reaches $46.345 \pm 1.288\%$ IoU and $63.330 \pm 1.197\%$ Dice, which is close to Conv-Adapter ($46.533 \pm 0.735\%$ IoU) and SSF ($46.442 \pm 1.361\%$ IoU), while using only 792 adapter-specific parameters. The total trainable count is dominated by the segmentation head: DT1D-Adapter uses 8,049,177 trainable parameters and 30.705 MB of FP32 task storage, compared with 11,020,337 parameters and 42.039 MB for Full fine-tuning. DT1D-Adapter does not provide a latency advantage in this rerun; its measured latency is 3.268 ± 0.349 ms/image compared with 2.525 ± 0.144 ms/image for Full fine-tuning.

The completed PennFudan ViT-B/16 dense-decoder rerun gives a different ranking from the DeepLabV3-MobileNetV3-Large experiment. BitFit obtains the highest supplied mean IoU at $66.874 \pm 1.558\%$, followed by DT1D-Adapter at $64.279 \pm 0.694\%$ and Linear probing at $63.557 \pm 0.916\%$. The supplied Full fine-tuning run reaches $39.001 \pm 0.991\%$ IoU and $56.111 \pm 1.026\%$ Dice, so it is retained as measured rather

Table 22: Binary mask segmentation accuracy on PennFudan using DeepLabV3-MobileNetV3-Large. Values are mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$. IoU gap is Full fine-tuning minus the corresponding method mean, in percentage points (pp).

Method	IoU (%)	Dice (%)	Pixel Acc. (%)	IoU Gap (pp)
DT1D-Adapter	46.345 $\pm$ 1.288	63.330 $\pm$ 1.197	86.574 $\pm$ 0.335	21.508
Full fine-tuning	67.853 $\pm$ 1.604	80.841 $\pm$ 1.145	93.293 $\pm$ 0.644	0.000
Linear probing	46.139 $\pm$ 1.790	63.130 $\pm$ 1.665	86.314 $\pm$ 0.661	21.715
BitFit [14]	44.012 $\pm$ 1.919	61.106 $\pm$ 1.841	85.693 $\pm$ 1.189	23.842
Conv-Adapter [36] ($r = 4$)	46.533 $\pm$ 0.735	63.509 $\pm$ 0.685	86.950 $\pm$ 0.499	21.321
Residual Adapters [38]	45.193 $\pm$ 2.022	62.235 $\pm$ 1.904	86.953 $\pm$ 0.300	22.660
SSF [16]	46.442 $\pm$ 1.361	63.419 $\pm$ 1.265	86.530 $\pm$ 0.557	21.412
BAM [42]	45.787 $\pm$ 0.486	62.813 $\pm$ 0.458	86.869 $\pm$ 0.443	22.066
LoRA-Conv [13, 43]	49.860 $\pm$ 1.334	66.535 $\pm$ 1.184	87.973 $\pm$ 0.747	17.994

Bold values in the PEFT rows identify the strongest supplied PEFT mean; Full fine-tuning is shown separately as the unconstrained reference.

Table 23: Binary mask segmentation accuracy on PennFudan using the ViT-B/16 dense decoder. Values are mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$. IoU gap is Full fine-tuning minus the corresponding method mean, in percentage points (pp).

Method	IoU (%)	Dice (%)	Pixel Acc. (%)	IoU Gap (pp)
DT1D-Adapter	64.279 $\pm$ 0.694	78.255 $\pm$ 0.513	92.509 $\pm$ 0.515	-25.278
Full fine-tuning	39.001 $\pm$ 0.991	56.111 $\pm$ 1.026	84.557 $\pm$ 1.268	0.000
Linear probing	63.557 $\pm$ 0.916	77.716 $\pm$ 0.683	92.341 $\pm$ 0.570	-24.556
BitFit [14]	66.874 $\pm$ 1.558	80.142 $\pm$ 1.125	93.160 $\pm$ 0.520	-27.873

Negative IoU-gap values indicate that the corresponding method exceeds the supplied Full fine-tuning mean. Bold values identify the strongest supplied mean among these four runs.

than treated as the best-accuracy reference. DT1D-Adapter uses 1,252 trainable parameters in total, of which 483 are adapter-specific, compared with 103,681 for BitFit and 85,799,425 for Full fine-tuning. Its mean latency is 12.556 ± 0.432 ms/image and its peak training memory is 406.207 MB, compared with 10.943 ± 0.128 ms/image and 1743.669 MB for Full fine-tuning.

5.2.6 Semantic segmentation

The completed D03 rerun provides three-seed semantic-segmentation results on Oxford-IIIT Pet using DeepLabV3-MobileNetV3. The five completed methods are DT1D-Adapter, Full fine-tuning, Linear probing, Conv-Adapter ($r = 4$), and BAM.

Table 24: Binary mask segmentation efficiency on PennFudan using DeepLabV3-MobileNetV3-Large [8, 60]. Values are mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$ where the measured quantity varies across seeds. Parameter reduction is computed from the total trainable parameter count relative to Full fine-tuning.

Method	Trainable params	Param. red. (%)	Storage (MB)	Latency (ms/img)	Peak Train Mem. (MB)	Epoch Time (s)
DT1D-Adapter	8,049,177	26.961	30.705	3.268 ± 0.349	326.125	2.229 ± 0.083
Full fine-tuning	11,020,337	0.000	42.039	2.525 ± 0.144	415.829	1.662 ± 0.344
Linear probing	8,048,385	26.968	30.702	2.905 ± 0.107	202.593	1.450 ± 0.033
BitFit [14]	8,065,665	26.811	30.768	2.635 ± 0.200	389.170	1.934 ± 0.053
Conv-Adapter [36] ($r = 4$)	8,528,779	22.609	32.535	2.796 ± 0.515	324.139	1.726 ± 0.283
Residual Adapters [38]	8,167,749	25.885	31.157	3.016 ± 0.323	317.533	1.713 ± 0.021
SSF [16]	8,050,865	26.945	30.712	2.555 ± 0.116	317.200	1.721 ± 0.029
BAM [42]	8,261,375	25.035	31.515	2.655 ± 0.185	323.500	1.780 ± 0.051
LoRA-Conv [13, 43]	8,142,593	26.113	31.062	3.864 ± 0.218	381.837	2.236 ± 0.100

The total trainable count includes the task head. Adapter-specific parameter counts are 792 for DT1D-Adapter, 2,480 for SSF, 94,208 for LoRA-Conv, 119,364 for Residual Adapters, 212,990 for BAM, and 480,394 for Conv-Adapter.

Table 25: Binary mask segmentation efficiency on PennFudan using the ViT-B/16 dense decoder [11]. Values are mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$ where the measured quantity varies across seeds. Parameter reduction is computed from the total trainable parameter count relative to Full fine-tuning.

Method	Trainable params	Param. red. (%)	Storage (MB)	Latency (ms/img)	Peak Train Mem. (MB)	Epoch Time (s)
DT1D-Adapter	1,252	99.999	0.005	12.556 ± 0.432	406.207	1.494 ± 0.009
Full fine-tuning	85,799,425	0.000	327.299	10.943 ± 0.128	1743.669	2.279 ± 0.041
Linear probing	769	99.999	0.003	13.955 ± 0.152	406.115	1.409 ± 0.039
BitFit [14]	103,681	99.879	0.396	10.966 ± 0.211	884.125	2.010 ± 0.025

DT1D-Adapter uses 483 adapter-specific parameters in addition to the 769 trainable dense-decoder/head parameters. Storage is the FP32 task-specific trainable-parameter footprint reported by the supplied rerun archives.

The completed D03 results show that Full fine-tuning obtains the highest mean semantic-segmentation accuracy at $77.231 \pm 0.089\%$ mIoU. Among the supplied PEFT methods, BAM has the highest mean mIoU at $65.475 \pm 0.590\%$, followed by DT1D-Adapter at $63.707 \pm 0.242\%$; Conv-Adapter and Linear probing reach $62.374 \pm 0.325\%$ and $62.298 \pm 0.206\%$, respectively. DT1D-Adapter uses only 792 adapter-specific

Table 26: Completed D03 semantic-segmentation accuracy and parameter results on Oxford-IIIT Pet using DeepLabV3-MobileNetV3 [28, 60]. All methods are trained for 5 epochs with input size 224×224 , batch size 8, and seeds $\{0, 1, 2\}$. For each seed, the test split is evaluated once at the checkpoint with the highest validation mIoU. Values that vary across seeds are reported as mean $\pm$ sample standard deviation.

Method	Total params	Trainable params	Trainable (%)	Best Epoch	mIoU (%)	Pixel Acc. (%)	Dice (%)
DT1D-Adapter	921,643	8,049,691	73.035	3.667 ± 0.577	63.707 ± 0.242	86.184 ± 0.159	74.105 ± 0.250
Full fine-tuning	11,020,851	11,020,851	100.000	3.667 ± 0.577	77.231 ± 0.089	91.997 ± 0.157	85.962 ± 0.027
Linear probing	11,020,851	8,048,899	73.033	3.667 ± 0.577	62.298 ± 0.206	85.547 ± 0.069	72.678 ± 0.302
Conv-Adapter [36] ($r = 4$)	1,501,245	8,529,293	74.160	4.000 ± 0.000	62.374 ± 0.325	85.569 ± 0.055	72.752 ± 0.398
BAM [23]	23,841	8,261,889	73.545	4.000 ± 0.000	65.475 ± 0.590	86.944 ± 0.324	75.827 ± 0.542

Bold values in the PEFT rows identify the strongest supplied PEFT mean; Full fine-tuning is shown separately as the unconstrained reference. Adapter-specific parameter counts are 792 for DT1D-Adapter, 212,990 for BAM, and 480,394 for Conv-Adapter.

Table 27: Completed D03 semantic-segmentation storage, runtime, throughput, and memory results on Oxford-IIIT Pet. Storage denotes the FP32 task-specific trainable-parameter footprint. Values that vary across seeds are reported as mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$.

Method	Storage (MB)	Latency (ms/img)	FPS	Peak Infer. Mem. (MB)	Peak Train Mem. (MB)	Epoch Time (s)	Train Time (s)
DT1D-Adapter	9.707	3.312 ± 0.177	302.493 ± 16.686	198.330	339.754	44.048 ± 3.164	249.605 ± 17.382
Full fine-tuning	42.041	2.521 ± 0.207	398.460 ± 31.342	232.278	430.855	31.611 ± 7.511	180.966 ± 43.049
Linear probing	30.704	2.621 ± 0.133	382.158 ± 19.351	198.999	207.195	21.721 ± 1.080	132.108 ± 6.489
Conv-Adapter [36] ($r = 4$)	32.537	3.559 ± 0.898	291.810 ± 64.546	205.163	334.014	31.304 ± 1.277	182.848 ± 5.718
BAM [23]	0.117	3.076 ± 0.363	328.208 ± 39.974	202.011	334.337	31.679 ± 2.501	185.161 ± 13.040

parameters, compared with 212,990 for BAM and 480,394 for Conv-Adapter. Its measured latency is 3.312 ± 0.177 ms/image, so this rerun supports adapter-parameter compactness rather than a general latency advantage.

The completed D07 rerun adds an Oxford-IIIT Pet semantic-segmentation comparison with a ViT-B/16 dense decoder under a common 5-epoch training budget. DT1D-Adapter reaches $73.634 \pm 0.342\%$ mIoU, $90.503 \pm 0.043\%$ pixel accuracy, and $83.058 \pm 0.352\%$ Dice. BitFit obtains the strongest supplied mean mIoU at $76.671 \pm 0.246\%$, while Linear probing reaches $72.635 \pm 0.035\%$. The supplied Full fine-tuning run reaches $50.406 \pm 0.374\%$ mIoU under the same training budget and is retained as measured. DT1D-Adapter uses 2,790 trainable parameters in total, including 483 adapter-specific parameters and 2,307 trainable dense-decoder/head parameters.

Table 28: Completed D07 semantic-segmentation accuracy and parameter results on Oxford-IIIT Pet using a ViT-B/16 dense decoder [11]. All methods are trained for 5 epochs with input size 224×224 , batch size 8, and seeds $\{0, 1, 2\}$. For each seed, the test split is evaluated once at the checkpoint with the highest validation mIoU. Values that vary across seeds are reported as mean $\pm$ sample standard deviation.

Method	Total params	Trainable params	Trainable (%)	Best Epoch	mIoU (%)	Pixel Acc. (%)	Dice (%)	Test loss
DT1D-Adapter	85,801,446	2,790	0.003	3.000 ± 1.000	73.634 ± 0.342	90.503 ± 0.043	83.058 ± 0.352	0.242 ± 0.002
Full fine-tuning	85,800,963	85,800,963	100.000	4.000 ± 0.000	50.406 ± 0.374	78.964 ± 0.167	61.435 ± 0.465	0.528 ± 0.003
Linear probing	85,800,963	2,307	0.003	2.667 ± 1.155	72.635 ± 0.035	90.128 ± 0.054	82.244 ± 0.061	0.252 ± 0.002
BitFit [14]	85,800,963	105,219	0.123	3.000 ± 1.000	76.671 ± 0.246	91.876 ± 0.042	85.446 ± 0.211	0.213 ± 0.001

Bold values identify the strongest supplied mean among the four completed D07 methods. DT1D-Adapter uses 483 adapter-specific parameters in addition to 2,307 trainable dense-decoder/head parameters.

Table 29: Completed D07 semantic-segmentation storage, runtime, throughput, and memory results on Oxford-IIIT Pet using the ViT-B/16 dense decoder. Storage denotes the FP32 task-specific trainable-parameter footprint. Values that vary across seeds are reported as mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$.

Method	Storage (MB)	Latency (ms/img)	FPS	Peak infer. mem. (MB)	Peak train mem. (MB)	Epoch time (s)	Train time (s)
DT1D-Adapter	0.011	13.424 ± 0.228	74.510 ± 1.276	401.167	408.824	25.256 ± 0.178	179.726 ± 4.183
Full fine-tuning	327.305	12.576 ± 0.065	79.519 ± 0.408	1391.088	1746.093	47.469 ± 0.125	288.275 ± 0.579
Linear probing	0.009	12.757 ± 0.193	78.400 ± 1.182	401.075	408.732	21.800 ± 0.384	161.927 ± 2.139
BitFit [14]	0.401	13.376 ± 0.053	74.762 ± 0.296	410.378	898.365	42.763 ± 0.047	267.285 ± 0.629

The D07 efficiency profile reinforces the parameter/storage interpretation of DT1D-Adapter. Its FP32 task-specific footprint is approximately 0.011 MB, compared with 0.401 MB for BitFit and 327.305 MB for Full fine-tuning. Its 13.424 ± 0.228 ms/image latency is similar to the other ViT dense-decoder methods, while peak training memory is 408.824 MB, close to Linear probing and substantially below BitFit and Full fine-tuning. These results support a compact task-specific adaptation footprint, while BitFit remains stronger in raw mIoU under the present D07 training protocol.

5.2.7 Object detection

The updated D06 10-epoch rerun provides three-seed object-detection results on Oxford-IIIT Pet using Faster R-CNN with a MobileNetV3-FPN backbone. DT1D-Adapter reaches $96.940 \pm 0.359\%$ AP₅₀ and $97.911 \pm 0.713\%$ Recall₅₀. BAM obtains the highest mean AP₅₀ at $98.191 \pm 0.377\%$, while DT1D-Adapter remains close and adds only 792 adapter-specific parameters, compared with 212,990 for BAM and 480,394 for Conv-Adapter. Conv-Adapter reaches $96.820 \pm 0.405\%$ AP₅₀ and the existing Head-only baseline (the detection-pipeline equivalent of the common dense Linear/Head-only baseline) reaches $95.593 \pm 1.390\%$. The supplied Full fine-tuning aggregate remains unstable after the longer training budget: seeds 1 and 2 yield zero AP₅₀ and zero Recall₅₀, while seed 0 yields nonzero detection performance. We therefore retain all three supplied runs in the aggregate and do not use this Full fine-tuning result as evidence of a general disadvantage.

Table 30: Three-seed object-detection accuracy and parameter results on Oxford-IIIT Pet using Faster R-CNN with MobileNetV3-FPN. All methods are trained for 10 epochs with input size 320×320 , batch size 1, GPU execution, and seeds $\{0, 1, 2\}$. Accuracy and count values are mean $\pm$ sample standard deviation where they vary across seeds. Adapter parameters denote the method-specific adaptation parameters in addition to the trainable detector head or other task-specific components.

Method	AP ₅₀ (%)	Recall ₅₀ (%)	GT	Pred.	Total params	Trainable params	Adapter params	Trainable (%)
DT1D-Adapter	96.940 ± 0.359	97.911 ± 0.713	79.667 ± 0.577	107.000 ± 8.718	18,931,021	14,516,077	792	76.679
Full fine-tuning	4.665 ± 8.081	32.083 ± 55.570	79.667 ± 0.577	194.333 ± 336.595	18,930,229	18,930,229	0	100.000
Linear / Head-only	95.593 ± 1.390	96.661 ± 1.901	79.667 ± 0.577	104.000 ± 14.933	18,930,229	14,515,285	0	76.678
Conv-Adapter [36] ($r = 4$)	96.820 ± 0.405	97.911 ± 0.713	79.667 ± 0.577	113.000 ± 18.358	19,410,623	14,995,679	480,394	77.255
BAM [42]	98.191 ± 0.377	98.745 ± 0.009	79.667 ± 0.577	100.000 ± 8.888	19,143,219	14,728,275	212,990	76.937

The Full fine-tuning aggregate includes all supplied runs; seeds 1 and 2 have zero AP₅₀ and zero Recall₅₀ and were not manually excluded.

Table 31: Three-seed object-detection storage, runtime, throughput, and memory results on Oxford-IIIT Pet. Storage denotes FP32 task-specific trainable-parameter storage. Values that vary across seeds are reported as mean $\pm$ sample standard deviation.

Method	Storage (MB)	Latency (ms/img)	FPS	Peak infer. mem. (MB)	Peak train mem. (MB)	Epoch time (s)	Train time (s)
DT1D-Adapter	55.374	27.557 ± 1.610	36.372 ± 2.160	278.656	1880.202	323.355 ± 6.525	3541.536 ± 72.095
Full fine-tuning	72.213	20.426 ± 3.649	49.952 ± 8.370	329.040	1938.111	225.272 ± 5.728	2470.587 ± 58.916
Linear / Head-only	55.371	22.855 ± 3.262	44.316 ± 5.921	278.483	312.377	158.838 ± 2.482	1850.673 ± 31.780
Conv-Adapter [36] ($r = 4$)	57.204	24.130 ± 0.365	41.448 ± 0.629	285.829	1884.856	295.413 ± 5.361	3239.717 ± 61.451
BAM [42]	56.184	23.785 ± 0.597	42.060 ± 1.063	281.802	1882.827	262.784 ± 12.754	2897.907 ± 144.914

The supplied F04 archive contains a current single-seed accuracy-parameter trade-off figure for the completed FGVC-Aircraft/ResNet-18 10-epoch reruns.

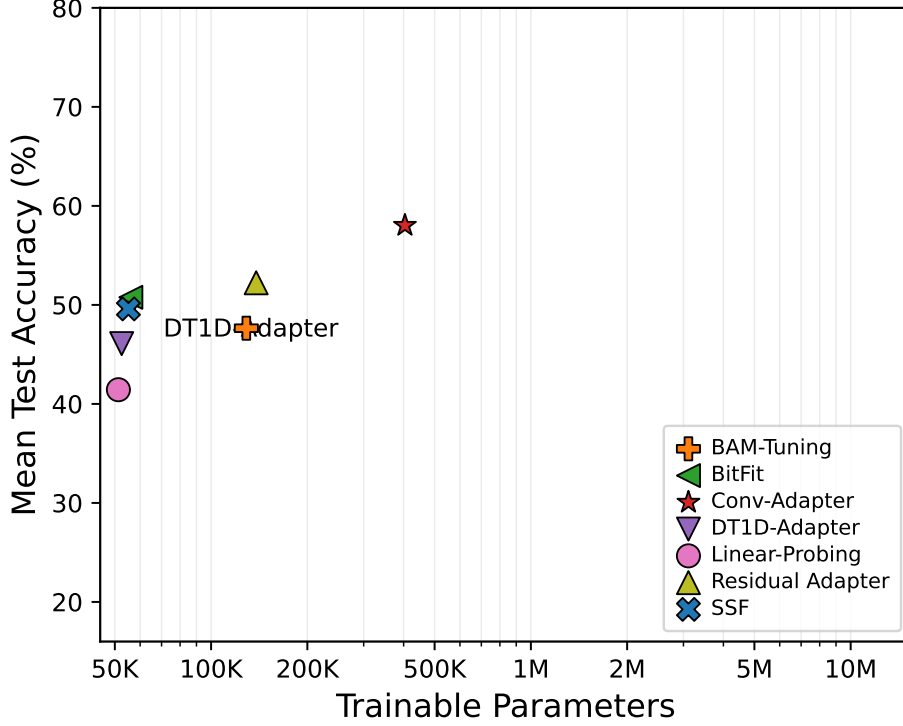


Fig. 4: Current accuracy–parameter trade-off on FGVC-Aircraft with an ImageNet-pretrained ResNet-18 backbone trained for 10 epochs, using representative seed 0. The figure includes the seven completed F04 methods available in the supplied archive; current Full fine-tuning, LoRA-Conv, and Side-Tuning runs are not available and are therefore not shown.

6 Conclusion

This paper introduced **DT1D-Adapter**, a structured parameter-efficient spatial adaptation module for visual backbones. The final formulation represents each group of direct axial kernels using a group-shared symmetric base response on radii $\{1, 2, 4\}$ and a normalized radius-4 channel-dependent correction. Each axis then learns a bounded $p = 2$ symmetric coefficient shift, initialized at zero and fused into an effective 13-tap kernel. The shifted height and width kernels are jointly normalized, executed through one depthwise convolution per axis, and injected through a learned residual gate.

The ablation results support the parameter-compact design. On DTD/ResNet-18, the final method obtains $62.163 \pm 0.406\%$ test Acc@1 with 25,335 trainable parameters, compared with $61.418 \pm 0.374\%$ and 89,399 parameters for the unrestricted axial depthwise control. On Flowers102/ResNet-50, the corresponding values are $81.477 \pm 0.551\%$ with 218,486 parameters versus $80.577 \pm 0.277\%$ with 722,550 parameters. The

learned weighted-shift increment is modest in both settings, whereas removing the post-shift joint ℓ_1 projection produces the larger accuracy decrease.

The currently complete ViT reruns show that the same spatial interface can be used with Transformer backbones, while also revealing clear limitation cases. On Flowers102 with ViT-B/16, DT1D-Adapter reaches $96.092 \pm 0.203\%$ test Acc@1 at batch size 32 and $95.864 \pm 0.191\%$ at batch size 16, using 84,234 trainable parameters. The batch-size-32 result exceeds the supplied Full fine-tuning, VPT, and Linear probing means while remaining below Pfeiffer Adapter ($96.867 \pm 0.047\%$), which uses 972,966 trainable parameters. On VTAB-Caltech101, DT1D-Adapter reaches $82.249 \pm 1.078\%$ with 84,234 trainable parameters, closely matching VPT at $82.495 \pm 1.335\%$ while using slightly fewer trainable parameters. On VTAB-DTD it reaches $49.450 \pm 1.413\%$ with 41,939 trainable parameters, and on VTAB-EuroSAT it reaches $77.179 \pm 0.593\%$ with 13,486 trainable parameters. In the latter two settings, stronger raw accuracy is obtained by other supplied baselines. Accordingly, DT1D-Adapter is positioned as a compact structured spatial PEFT alternative rather than a universally dominant method.

The completed PennFudan binary-segmentation reruns provide complementary dense-prediction results. With DeepLabV3-MobileNetV3-Large, DT1D-Adapter obtains $46.345 \pm 1.288\%$ foreground IoU and $63.330 \pm 1.197\%$ Dice over three seeds. LoRA-Conv is the strongest supplied PEFT baseline in that experiment at $49.860 \pm 1.334\%$ IoU, while Full fine-tuning reaches $67.853 \pm 1.604\%$. DT1D-Adapter nevertheless uses only 792 adapter-specific parameters, compared with 94,208 for LoRA-Conv, 119,364 for Residual Adapters, 212,990 for BAM, and 480,394 for Conv-Adapter. With the ViT-B/16 dense decoder, DT1D-Adapter reaches $64.279 \pm 0.694\%$ IoU and $78.255 \pm 0.513\%$ Dice using 1,252 trainable parameters in total, including 483 adapter-specific parameters. BitFit is strongest among the four supplied ViT dense runs at $66.874 \pm 1.558\%$ IoU, whereas the supplied Full fine-tuning run reaches $39.001 \pm 0.991\%$. These segmentation results support the compactness of the proposed spatial branch but do not support a claim of universally best raw segmentation accuracy.

The completed D03 semantic-segmentation rerun on Oxford-IIIT Pet provides an additional dense-prediction comparison. Full fine-tuning reaches $77.231 \pm 0.089\%$ mIoU, while BAM is the strongest supplied PEFT method at $65.475 \pm 0.590\%$. DT1D-Adapter reaches $63.707 \pm 0.242\%$ mIoU and $74.105 \pm 0.250\%$ Dice with only 792 adapter-specific parameters, compared with 212,990 for BAM and 480,394 for Conv-Adapter. Its latency is 3.312 ± 0.177 ms/image, so the D03 result again supports parameter compactness without establishing a universal runtime advantage.

The completed D07 Oxford-IIIT Pet ViT-B/16 semantic-segmentation rerun further tests the same spatial interface with a Transformer backbone. DT1D-Adapter reaches $73.634 \pm 0.342\%$ mIoU and $83.058 \pm 0.352\%$ Dice with 2,790 trainable parameters in total, including 483 adapter-specific parameters. BitFit obtains the strongest supplied mean mIoU at $76.671 \pm 0.246\%$ with 105,219 trainable parameters, while Linear probing reaches $72.635 \pm 0.035\%$. DT1D-Adapter therefore provides a favorable parameter footprint while remaining below BitFit in raw mIoU under this 5-epoch protocol.

The updated 10-epoch object-detection rerun shows that DT1D-Adapter reaches $96.940 \pm 0.359\%$ AP₅₀ and $97.911 \pm 0.713\%$ Recall₅₀ on Oxford-IIIT Pet with Faster R-CNN/MobileNetV3-FPN. BAM obtains the strongest supplied mean AP₅₀ at $98.191 \pm$

0.377%, while DT1D-Adapter uses only 792 adapter-specific parameters compared with 212,990 for BAM and 480,394 for Conv-Adapter. The supplied Full fine-tuning detection aggregate remains unstable because two seeds yield zero AP_{50} and zero $Recall_{50}$; all supplied runs are retained rather than discarded.

The newly incorporated CNN efficiency profiles reinforce the distinction between parameter efficiency and wall-clock efficiency. On Flowers102/ResNet-18 at 10 epochs, DT1D-Adapter reaches $83.011 \pm 0.244\%$ test $Acc@1$ with 53,550 trainable parameters, close to BitFit ($83.157 \pm 0.474\%$) and SSF ($83.103 \pm 0.818\%$), while Full fine-tuning reaches $88.415 \pm 0.206\%$. The corresponding C03 profiler reports 3.686 G FLOPs and 1.770 ± 0.117 ms/image for DT1D-Adapter, versus 1.819 G FLOPs and sub-0.60 ms/image latency for the supplied Linear, BitFit, and SSF profiles. On Oxford-IIIT Pet/EfficientNet-B0 at 10 epochs, DT1D-Adapter uses only 48,595 trainable parameters and about 0.185 MiB of FP32 task-specific storage while reaching $91.905 \pm 0.109\%$ test $Acc@1$, but its measured 3.995 ± 0.466 ms/image latency is higher than the completed Linear, BitFit, SSF, and Conv-Adapter profiles. On EuroSAT/MobileNetV3-Small, the currently complete C13 Linear-probing row provides a reference profile of 0.758 ± 0.010 ms/image and 1319.887 ± 17.330 FPS. These measurements are therefore reported alongside accuracy and parameter counts rather than treating parameter count as a proxy for runtime efficiency.

Several CNN and other result reruns are still pending in this working manuscript version. In particular, C11 and C13 remain partial beyond the completed rows reported above. Their unavailable table cells remain intentionally blank rather than carrying forward legacy values. The final quantitative discussion should be updated only after the corresponding current reruns are available.

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Data Availability

No new dataset was collected or created in this study. The classification experiments use publicly available benchmarks, including DTD R1.0.1, Oxford Flowers102, Oxford-IIIT Pet, EuroSAT RGB, Caltech101, VTAB-Caltech101, VTAB-DTD, and VTAB-EuroSAT. DTD uses official partition 1, Flowers102 uses its official training, validation, and test splits, and the corrected VTAB-Caltech101, VTAB-DTD, and VTAB-EuroSAT evaluations follow the VTAB-1k protocol [72], using `train800` for training, `val200` for validation, and the official test set for testing. For datasets without a separate official validation set, the training and validation subsets were generated from the official training data according to the split protocol described in the experimental section. The dense-prediction experiments use the publicly available PennFudan Pedestrian and Oxford-IIIT Pet datasets.

Images were resized to the input resolution reported in the corresponding tables. Training-time augmentation and deterministic evaluation preprocessing were applied according to the released experiment configurations, with ImageNet normalization for ImageNet-pretrained backbones. The accompanying repository provides the source code, dataset preparation instructions, configuration files, random seeds, preprocessing settings, experiment commands, processed logs, and summary result files used to reproduce the reported tables and figures.

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